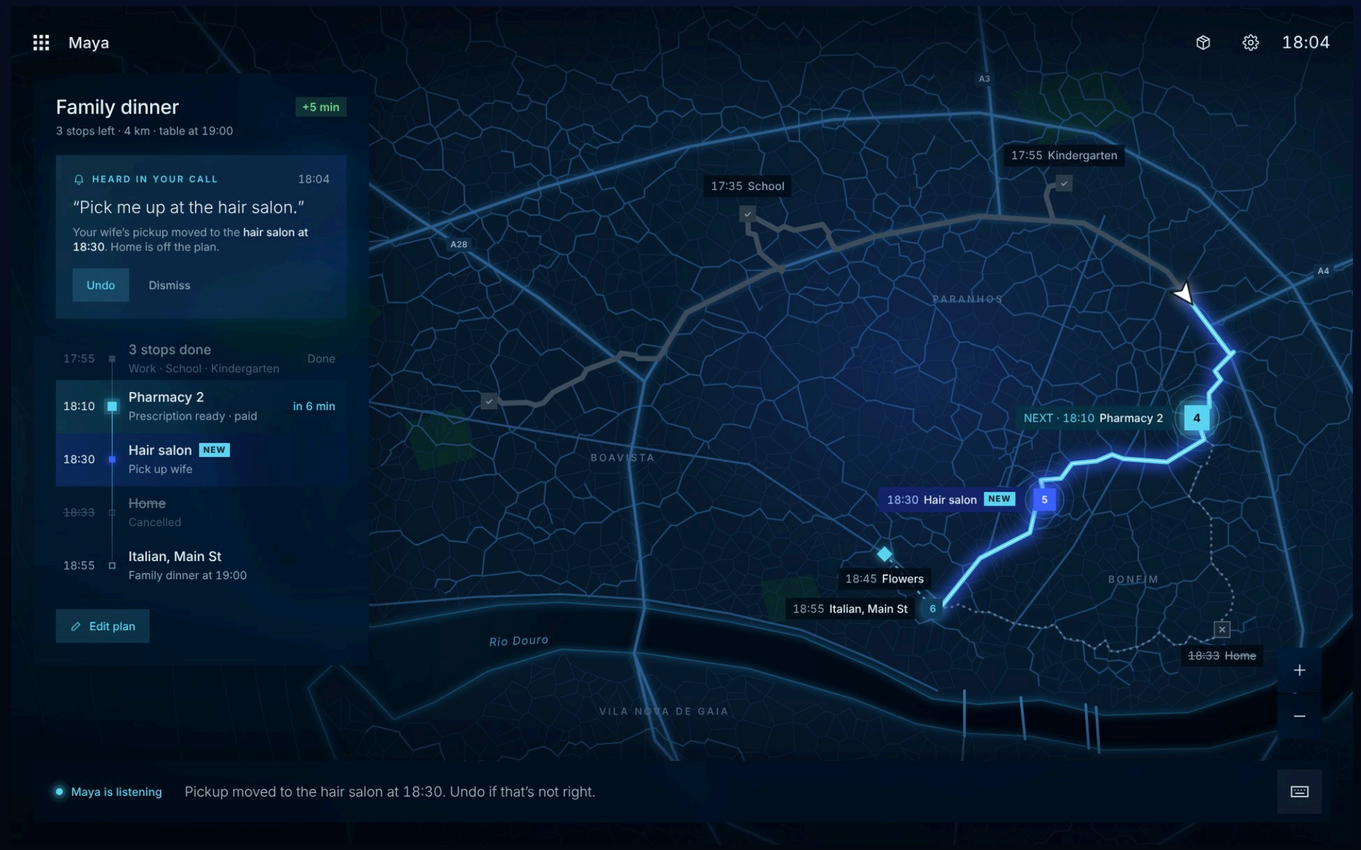
UX CHALLENGE · NEW AI ASSISTANT FOR INFOTAINMENT

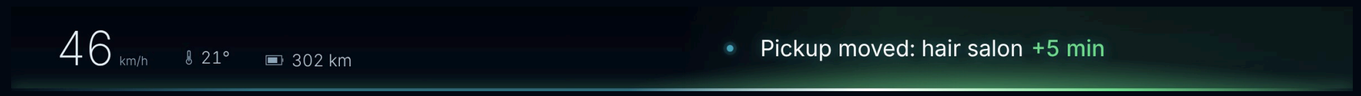
# Maya

## An AI trip concierge for any Level 5 car

Michael shares his needs and goals,
Maya plans the stops, phones the shops,
pays and adapts when conditions change.
When the car is free, she runs his errands.




CENTRAL DISPLAY




PANORAMIC VISION

**Aníbal Oliveira** · Senior Product Designer
October 2026

# In this document

Each of the seven sections below is a link.


01 **What I made, and why** — p. 3
The artifacts supporting this exercise, and why each one exists

02 **The idea** — p. 4–6
The problem and the bet · courier mode · principles and Level 5

03 **The journey** — p. 7–8
The scenario in ten beats, every row of the journey map

04 **The system** — p. 9–11
Surfaces and channels · screen anatomy · task flow

05 **Hi-fi design mock-ups** — p. 12–24
High-fidelity mock-ups supporting each scene, in the order of the story

06 **Design system** — p. 25–27
Foundations, components and more

07 **Rules and validation** — p. 28–30
Edge cases · validation and metrics · trade-offs and what’s next

# Each artifact answered one question before the next one started.

| | | |
|---|---|---|
| 01 | **Concept foundations** | To commit to one bet and rank what could break it, so the riskiest parts (ambient intent, AI callers, an empty car) get designed and tested first. |
| 02 | **Hero scenario and conversation script** | An assistant is mostly conversation. Writing every line first decided when Maya speaks, when she stays silent and what the screen must show instead. |
| 03 | **Journey map** | To see the whole evening at once: Michael's goals and feelings, what Maya does backstage and where the risks sit. |
| 04 | **Task flow** | The logic behind every replan and the courier hand-off, one decision at a time. |
| 05 | **Screen specs and mood board** | To fix the screen inventory and IDs, decide what Panoramic Vision may show, and choose a direction: Deep ocean, because in a dark cabin light can carry meaning. |
| 06 | **Multimodal matrix** | To keep voice and touch equal: every voice action has a touch twin, every spoken line a caption. |
| 07 | **Edge cases and validation plan** | To design for the evening going wrong, and to say how each assumption would be tested and what success looks like. |
| 08 | **Central display wireframes** | To settle layout and hierarchy (plan on the left, map behind, Maya below) before any visual decisions. |
| 09 | **Hi-fi screens and components** | The whole story in final quality, built from one set of components so every state stays consistent. |
| 10 | **Clickable prototype** | To feel the timing of the story, and to run the remote tests in the validation plan. |

THREE LINKS, ALL OPTIONAL

Plain web pages, no sign-in. This PDF stands on its own; the links add full-size screens, the working files and the prototype.

ENTRY PAGE

**oliveiraanibal.github.io/maya/**

This PDF and every working artifact, in the order they were made.

SCREENS

**oliveiraanibal.github.io/maya/screens/**

All hi-fi screens at full size, the Panoramic Vision states and the component inventory.

PROTOTYPE

**oliveiraanibal.github.io/maya/prototype/**

The whole evening by touch or voice, with edge cases 1 to 4 as what-if branches.

WORKING ALONE

A solo exercise with no user research. The assumptions are written down, ranked by risk, and paired with the tests that would check them.

# A self-driving car frees your hands, not your evening.

Navigation handles routes, not commitments. Even when the car drives, the passenger still works out the order of the stops, phones businesses and replans by hand whenever something changes, so the freed-up time is not really free.

INSIGHT

### People think in needs, not addresses

“Family dinner tonight. Let’s get the kids and my wife, and I need that new prescription.” One sentence holds three pickups, a pharmacy to find and pay, a restaurant and a deadline.

VISION

### “I share my needs and goals; everything else just happens, and I can see it happening.”

“With me in the car, Maya runs my trip. Without me, my car runs my errands.”

HYPOTHESIS

Time-poor people will hand trip logistics to an AI that **asks only what it cannot know**, acts on its own, follows the conversation and makes every action **visible and reversible**.

We are right when people arrive on time, rarely undo what Maya did, and trust her with more over time.

WHO

Michael, a father of two with a packed evening. Maya belongs to him, not to the car: she is in any car he signs in to, his own or a robotaxi.

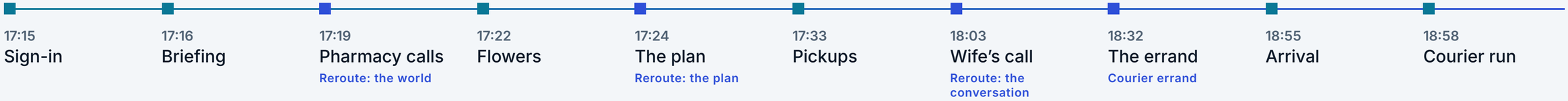
ASSUMED, NOT DESIGNED

Payments from Maya Wallet up to a limit he sets. Schools release the children to the car. The florist delivers to the restaurant.

ACCESSIBLE FROM THE START

Every task works with voice alone or touch alone. Everything Maya says is captioned, and colour never carries meaning on its own.

THE HERO SCENARIO: THE FAMILY DINNER RUN, ONE EVENING IN TEN SCENES



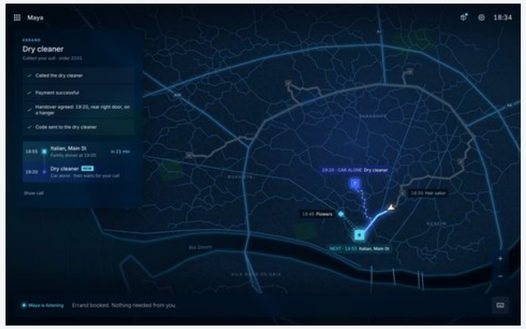
# Courier mode: when nobody is in the car, it runs errands on its own.

In the scenario, a message says Michael's suit is ready at the dry cleaner. Maya offers to collect it once the family is at the restaurant. Afterwards the car charges nearby and waits until Michael calls it, with the suit inside.



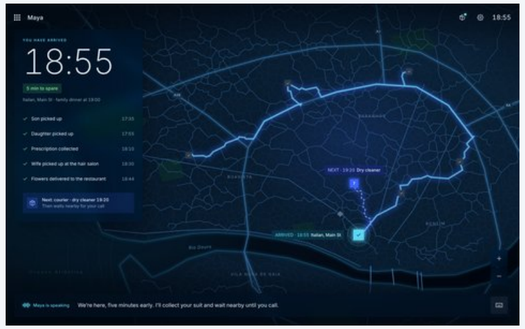

**S18 · Errand suggestion**

A message says the suit is ready. Maya offers to collect it with the empty car.



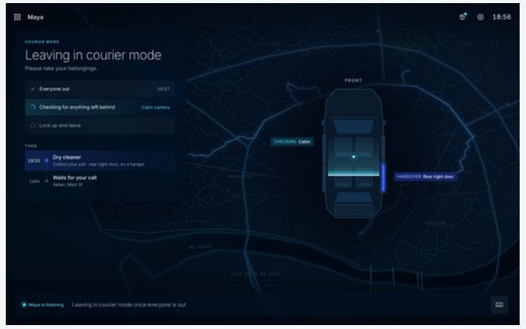

**S19 · Errand booked**

She calls, pays and agrees the handover. Only then is the errand booked.



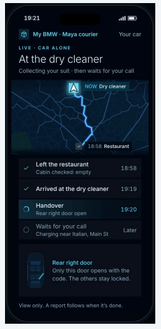

**S20 · Arrival report**

The arrival report ends with what happens next: the errand, then the car waits for his call.




**C1 · Leaving in courier mode**

The car checks that the cabin is empty and nothing is left behind.




**C2 · Phone: courier run**

Michael follows the run on his phone, view only.




**C3 · Phone: errand report**

A report with a photo, the receipt and the times.

## THE RULES

**Empty cabin only**
Before leaving, the car checks for people, pets and forgotten items.

**No handoff, no commitment**
An errand is booked only once the shop confirms the time, the person and the handover.

**One door, one code**
Only the agreed door opens, once, with a one-time code; the cabin camera checks what goes in.

**Ready when called**
After the errand the car charges nearby; when Michael calls it, the phone says how soon it will arrive.

**Every run is reported**
Live on the phone, then a photo, the receipt and the times.

**Charging is not an errand**
It never appears in the plan. The car charges out of sight.

### RISKIEST ASSUMPTIONS

A13 shops will hand items to an empty car · A14 people will send it, and accept a cabin camera · A15 an empty Level 5 car may legally run errands.

# Nine rules every screen follows, in a car nobody has to drive.

### 1 Ask only what she can't know

As many questions as needed, no more. Schools, pickups and the prescription come from the profile and messages.

S06

### 2 The plan follows the conversation

Ambient intent: what Maya hears in the cabin or a call can reshape the plan.

S17

### 3 Speak only when needed

The screen is the confirmation. Maya never reads the plan back.

S07 · S12

### 4 Do first, show clearly, undo easily

No approval gates within the limits he sets, and no countdowns. Every action lands in "Done for you".

S09 · S14 · S17

### 5 The plan is alive

The world, the conversation or Michael can reshape it, and the ripple is visible.

S09 · S15 · S17

### 6 Spare time is the promise

Always visible. Green above 4 min, amber 0 to 4, red below 0, never colour alone.

S12 to S20 · V2

### 7 Central display to act, Panoramic Vision to glance

Panoramic Vision can't be touched and carries nothing unique.

Panoramic Vision

### 8 Voice and touch are equals

Every task works with either one alone.

S05 · S06 · S15

### 9 With you or without you

An empty car works for you, but only with an agreed handover.

S18 to C3

## SAE DRIVING AUTOMATION LEVELS

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| No automation | Driver assistance | Partial | Conditional, takes over on request | High, in a defined area | **Full, anywhere, no driver** |

## WHY LEVEL 5

Only when nobody has to drive can an assistant use the whole trip, and can Michael talk, read and decide. One calm, trip-level experience instead of a design built around driver attention.

## AT LEVEL 3 OR 4

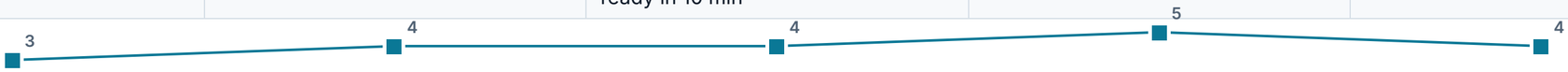
The same concept could ship earlier, but every task would give way to a takeover request: Maya pauses and goes silent, and the central display and Panoramic Vision count down together (V1, p. 28).

## HOW MICHAEL STAYS AWARE

- **The map is always there.** Position, the route ahead and what's done sit behind every screen.
- **The car states its intent first.** "Stop at School in 1 min", where it pulls in and which doors unlock (S13).
- **Every change shows what and why.** Removed stops crossed out, new ones marked, the old route dotted (S09, S17).
- **Car data has a fixed place.** Speed, temperature and range always hold the left half of Panoramic Vision.
- **Nothing happens out of sight.** Calls are transcribed, payments listed, recent actions can be undone.

# Setting off: from sign-in to a plan Michael can trust at a glance.

Emotion: Michael's expected feeling, 1 low to 5 high · A2 to A16: assumptions, p. 29

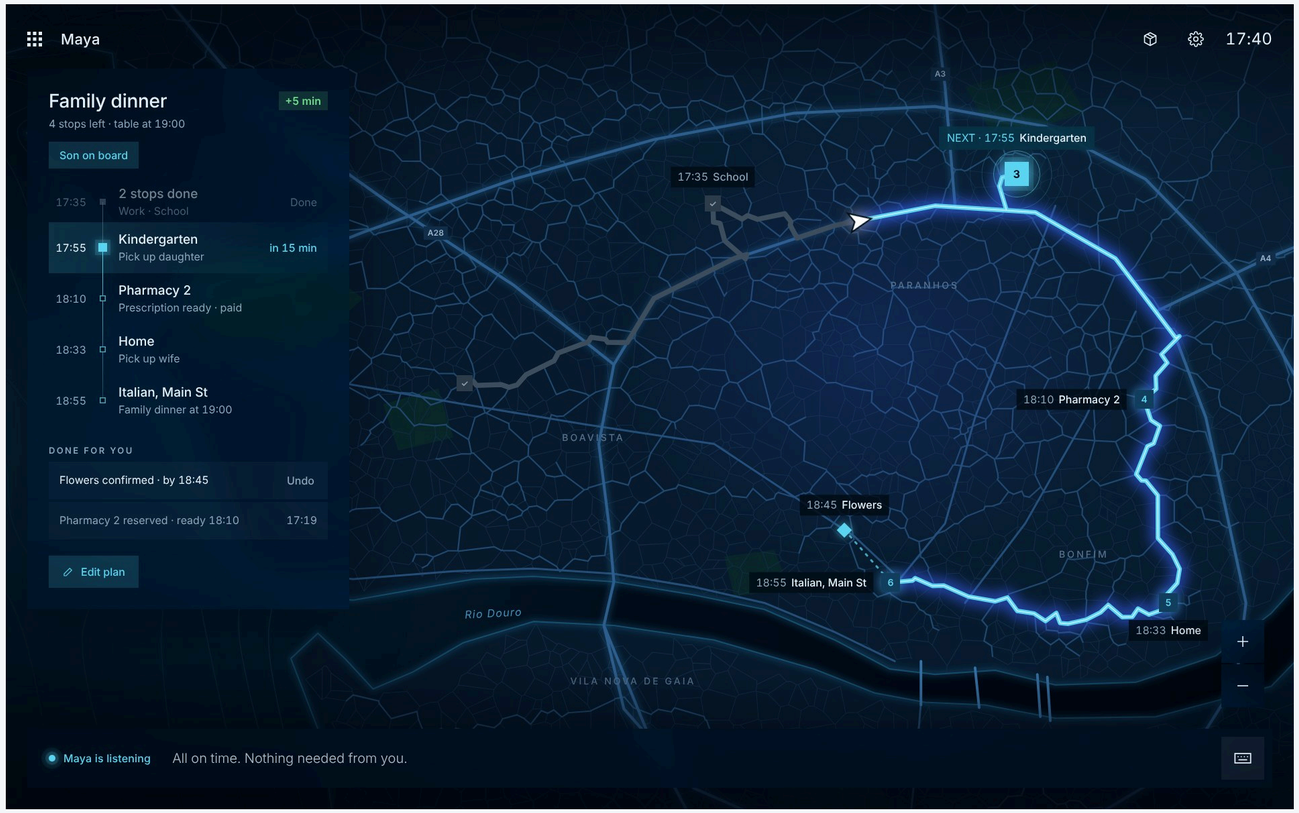
| PHASE | Setting off | | | | |
|---|---|---|---|---|---|
| TIME | 17:15 | 17:16 | 17:19 | 17:22 | 17:24 |
| BEAT | **Boarding and sign-in** | **Briefing** | **Pharmacy search** | **Flowers** | **The plan appears** |
| MICHAEL'S GOAL | Get going after work with no setup | Say what the evening needs, not where to go | Get the prescription as soon as possible | Not arrive empty-handed | Trust the plan at a glance |
| MICHAEL DOES OR SAYS | Gets in; signed in by face | "Family dinner tonight. Kids, my wife, and that prescription." Answers: the usual Italian on Main Street, at 19:00 | Rides; may read the transcripts | "Yes. White roses." | Glances at it |
| MAYA SAYS | "Hi, I'm Maya, your concierge for the trip. How can I assist you?" | Asks only where, and what time | Silent | Suggests flowers once the pharmacy is secured, then silent | Silent |
| CENTRAL DISPLAY SHOWS | Sign in, then Welcome and "Going home" (S01, S02) | The "New plan" card fills in (S04 to S07) | Two calls with holds, then "Plan updated" (S08, S09) | Suggestion, then call, payment and "Delivery committed" (S10, S11) | Plan overview, spare time +5 min (S12) |
| PANORAMIC VISION SHOWS | ◉ Home 17:48 | ◉ listening | Pharmacy changed · +5 min | ◉ Calling Main St Florist, then "Flowers confirmed" | School 17:35 · +5 min |
| MAYA BACKSTAGE | Recognises Michael; predicts home from habit; loads calendar, contacts, messages | Parses needs; reads the prescription from messages; asks the gaps | Calls pharmacies on the route; pays; reroutes to pharmacy 2 | Finds the birthday; picks an open florist near the restaurant; calls and pays from Maya Wallet | Builds the route; checks the range; arrival 18:55; replaces the default route |
| EXTERNAL | none | none | Pharmacy 1: no stock. Pharmacy 2: ready in 10 min | Florist agrees delivery by 18:45 | none |
| EMOTION |  | | | | |
| RISK | Wrong destination guess (A2); face not recognised | Misheard or missing detail (A6) | No pharmacy has stock (A4) | Florist refuses an AI caller (A4, A7) | Too much detail (A8) |
| DESIGN RESPONSE | The guess is a quiet default; PIN or password as fallback | The card is the confirmation; tap Maya's bar to type | Silent reroute with a visible banner | Self-introduction and live transcript | Spare time first, details on tap |

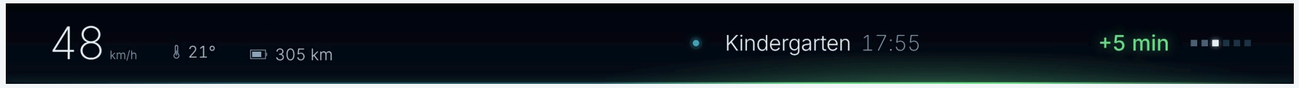
# On the way, arrival, and the car alone.

The last three rows are the service blueprint: backstage, the outside world, and how the design answers each risk.

| PHASE | On the way | | | Arrival | Car alone |
|---|---|---|---|---|---|
| TIME | 17:33 | 18:03 | 18:32 | 18:55 | 18:58 |
| BEAT | **Pickups** | **Wife’s call** | **The errand** | **Arrival** | **Courier run** |
| MICHAEL’S GOAL | Collect the kids without managing it | Absorb a change while on the phone | Get the suit without another trip | Arrive on time | Know the errand is done without watching it |
| MICHAEL DOES OR SAYS | Rides | Takes the call: “Sure, we’ll be there.” | “Yes, please.” | Steps out with the family | At dinner; glances at the phone |
| MAYA SAYS | Silent | Silent, chime only | Offers to collect the suit with the empty car after the drop-off | “We’re here, five minutes early. I’ll collect your suit and wait nearby until you call.” | Silent; notifications only |
| CENTRAL DISPLAY SHOWS | What the car is about to do, pickup states, “Done for you” (S13, S14) | Heard phrase, crossed and new stops, Undo (S16, S17) | Errand suggestion, then call, payment, agreed handover and a new plan row (S18, S19) | Arrival report, then “Leaving in courier mode” (S20, C1) | Central display off; the phone shows the live run and the report (C2, C3) |
| PANORAMIC VISION SHOWS | Next stop updates | Pickup moved: hair salon · +5 min | ◉ Calling the dry cleaner, then “Errand booked: dry cleaner” | Arrived 18:55 · +5 min, then “Courier · waits for you” | Off (cabin empty) |
| MAYA BACKSTAGE | Tracks pickups and ETAs; tells the car where to stop and which doors to unlock | Hears the call; infers the change; replans | Reads the dry cleaner’s message; checks the car is free after 18:55; calls, pays and agrees the handover; texts a one-time code | Confirms arrival; checks the cabin is empty and nothing is left behind | Drives to the dry cleaner; opens one door for the code; the camera checks the suit; reports; charges near the restaurant until Michael calls |
| EXTERNAL | School and kindergarten release the children | Wife asks to be picked up at the hair salon | Dry cleaner agrees to hang the suit in the rear right door at 19:20 | Flowers delivered 18:44; prescription collected 18:10 | Staff type the code and hang the suit at 19:20 |
| EMOTION |  | | | | |
| RISK | Child not at the pickup point | Ambient false positive (A3, A9) | Shops won’t serve an empty car (A13); trust in sending it (A14) | Someone or something left in the cabin (A14) | Shop not ready or wrong item (A13, A14) |
| DESIGN RESPONSE | The intent card says how long the car waits | Chime, heard phrase, Undo with no countdown | No handoff, no commitment: booked only once the handover is agreed | Courier mode starts only with an empty cabin | One door, one code; a report with photo, receipt and times |

# A central display to act, Panoramic Vision to glance, and the phone only when the car is alone.




### CENTRAL DISPLAY · ABOUT 15″, 1920 × 1200

Everything interactive. 16:10, seen from about 70 cm. The plan on the left, the live map behind, Maya's bar at the bottom.

### PANORAMIC VISION · BASE OF THE WINDSCREEN

Passive and always dark. Car data on the left, Maya on the right. A copy of the central display, never the only channel. Off when the cabin is empty.

## CHANNELS

**V — Voice**
Always listening in the cabin, no wake word. Only Michael's voice gives commands; other voices are context.

**T — Touch**
Everything on the central display: tap, drag, type. Every tap area is at least 76 × 76 px.

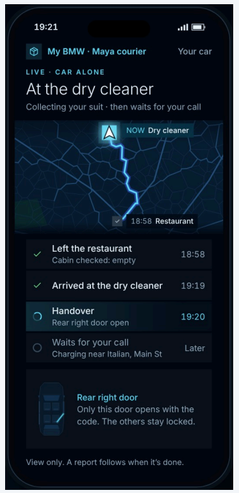
**T — Keyboard**
Tap Maya's bar to type. A large keyboard takes the lower half of the screen.

**S — Captions**
Everything Maya says appears in her bar, so nothing is audio only.

**S — Chime**
One soft sound, only for changes heard in conversation, always with the heard phrase on screen.

**S — Light**
The horizon line of Panoramic Vision carries the spare-time colour and flashes for 4 s on events.

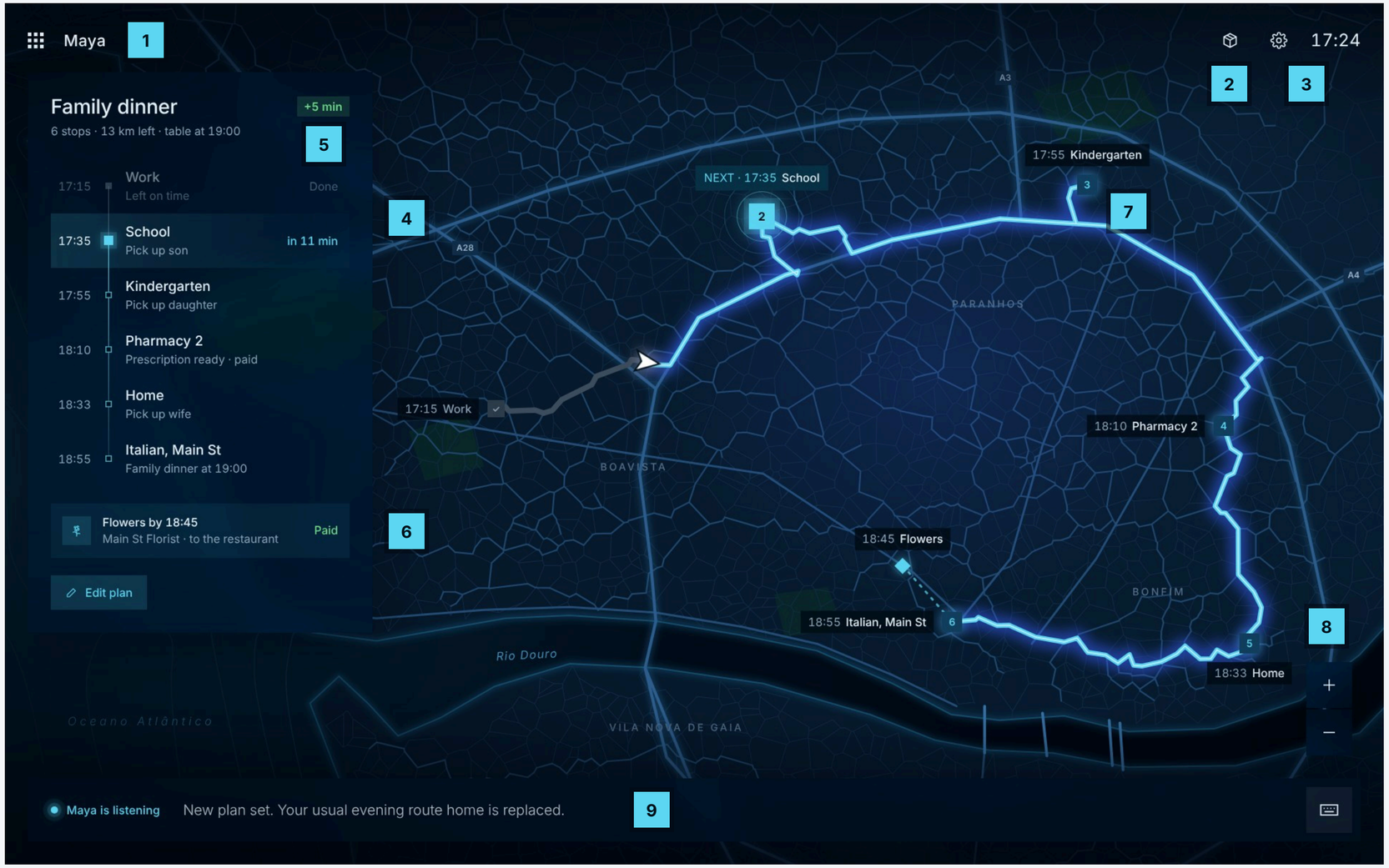
V voice  T touch  S system output




### PHONE

Courier mode only, and view only, in the My BMW app: the live run (C2) and the report (C3).

# One layout everywhere: the plan on the left, the map behind, Maya below.




**1 App launcher and Maya**
Other apps sit behind the launcher. “Maya” names what is on screen.

**2 Errands**
Opens Errands (C4). A dot shows while an errand is booked or running.

**3 Settings and clock**
Nothing else in the top bar.

**4 Plan panel**
Trip title, stops left, distance and the table time; the stop timeline, next stop highlighted; Edit plan. Glass over the map.

**5 Spare time**
The promise, always visible: +5 min in green.

**6 Done for you**
What Maya arranged: flowers by 18:45, paid.

**7 Live map**
The whole trip. The route ahead glows, the travelled route is grey; square pins; sonar rings on the next stop.

**8 Zoom**
In and out, aligned with the keyboard button.

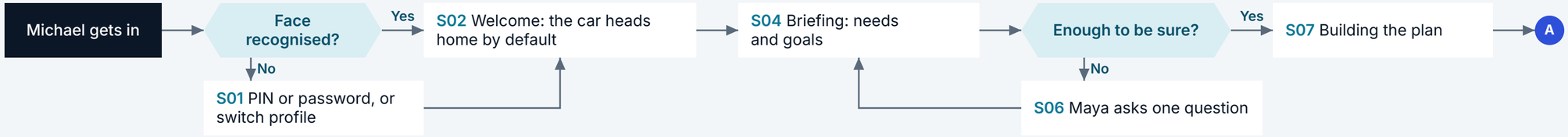
**9 Maya’s bar**
Her status (listening, thinking, speaking, on a call), her words as a caption, and the keyboard.

The map is Porto, drawn from its real coast, river and main roads; the minor streets are illustrative.

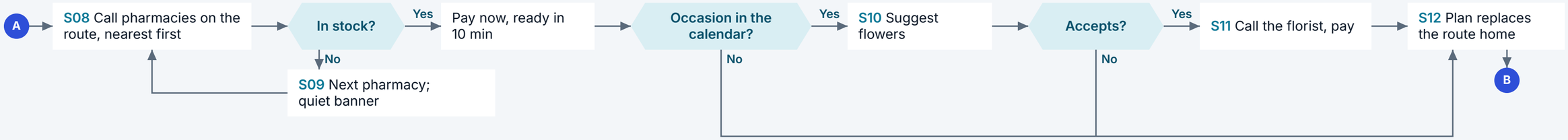
# Every decision behind the evening, in four lanes.

step · decision · continues · back to the trip

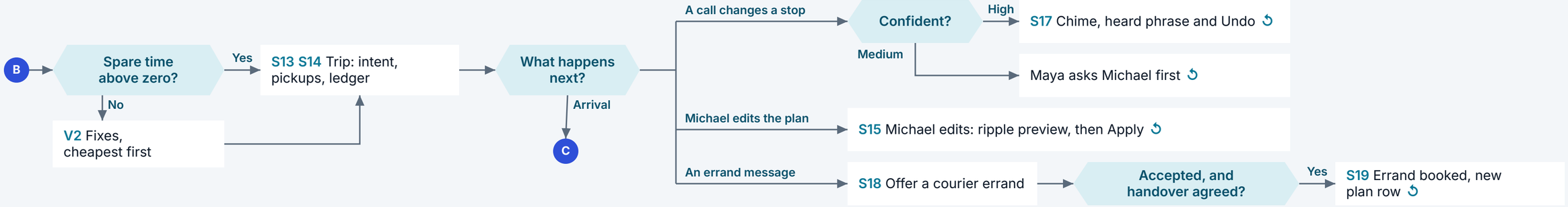
## 1 · GETTING GOING



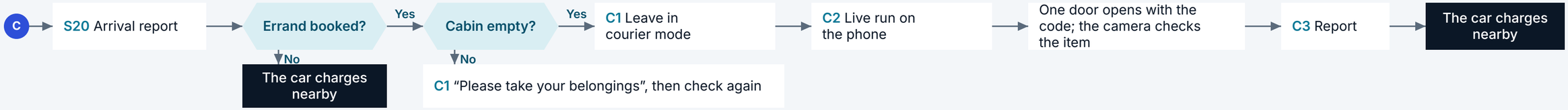

## 2 · PLANNING




## 3 · ON THE WAY

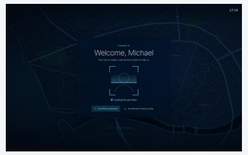



## 4 · AFTER ARRIVAL



# Every screen, in the order of the story. Each one is a link to its page.

S01 to S20 with Michael in the car, C1 to C5 for courier mode and errands, V1 and V2 for variants.




**S01** p. 13
Sign in



**S02** p. 13
Welcome



**S03** p. 13
Maya's access



**S04** p. 14
Briefing: listening



**S05** p. 14
Briefing: keyboard



**S06** p. 14
Briefing: Maya asks



**S07** p. 14
Briefing: building the plan



**S08** p. 15
Retailer calls



**S09** p. 16
Plan change: automatic



**S10** p. 17
Gift suggestion



**S11** p. 17
Florist order



**S12** p. 18
Plan overview



**S13** p. 19
Vehicle intent



**S14** p. 19
Active trip



**S15** p. 19
Edit plan



**S16** p. 20
Incoming call



**S17** p. 20
Plan change: ambient



**S18** p. 21
Errand suggestion



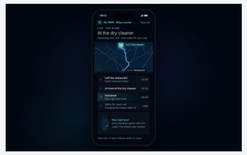
**S19** p. 21
Errand booked




**S20** p. 22
Arrival report




**C1** p. 22
Leaving in courier mode



**C2** p. 23
Phone: courier run



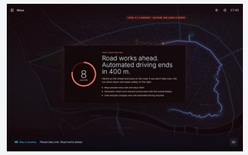
**C3** p. 23
Phone: errand report



**C4** p. 24
Errands



**C5** p. 24
New errand

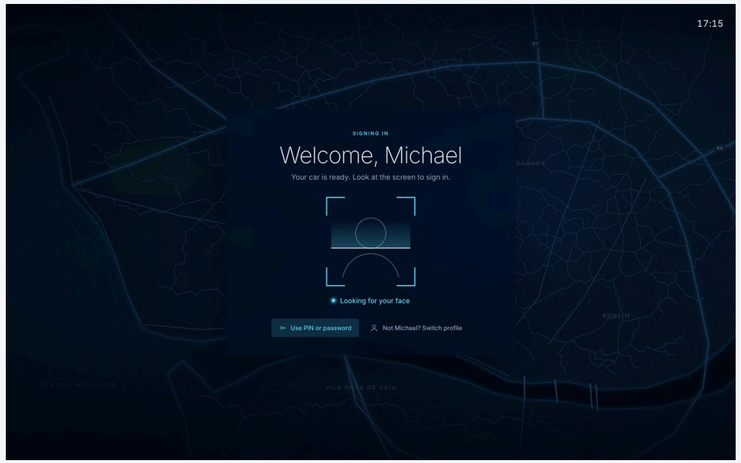



**V1** p. 28
Takeover request



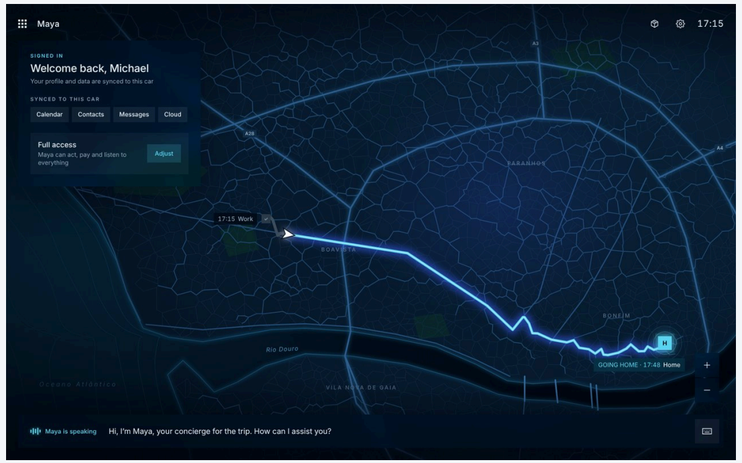
**V2** p. 28
Running late

# The car expects Michael, signs him in by face and sets off towards home.



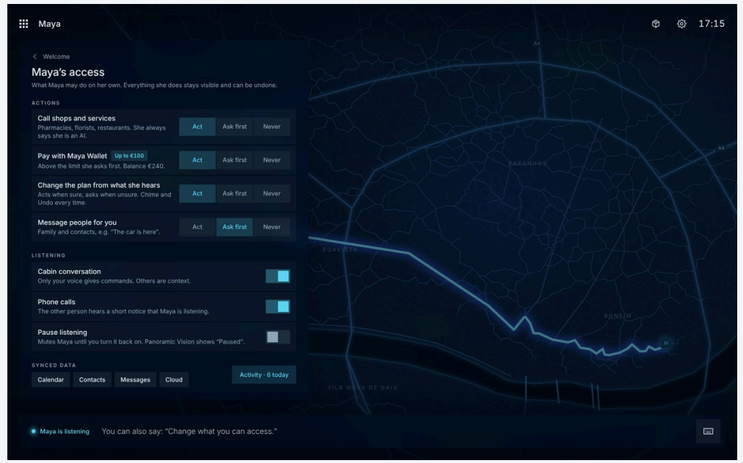

## S01 Sign in

- The car expects Michael but signs him in only once it recognises his face.
- Fallbacks: Use PIN or password, or “Not Michael? Switch profile”.
- Before sign-in the top bar shows only the clock.




## S02 Welcome

- “Welcome back, Michael”: calendar, contacts, messages and cloud synced to this car.
- Already moving on the most probable route: Going home, 17:48.
- “Full access” sums up Maya’s rights, with Adjust.




## S03 Maya’s access

- Opened from Adjust. Per action: Act, Ask first or Never.
- Maya Wallet pays on its own up to €100; above that, Maya asks first.
- Listening: cabin, phone calls (the other person hears a notice), pause.

### SPOKEN

| | |
|---|---|
| MAYA | “Hi, I’m Maya, your concierge for the trip. How can I assist you?” |
| TOUCH TWIN | Adjust opens Maya’s access; by voice, “Change what you can access.” |

### PANORAMIC VISION · MAYA’S HALF




Home 17:48. No plan yet, so spare time is not shown and the line stays cyan.

# Michael says what the evening needs. Maya asks only what she can’t know.

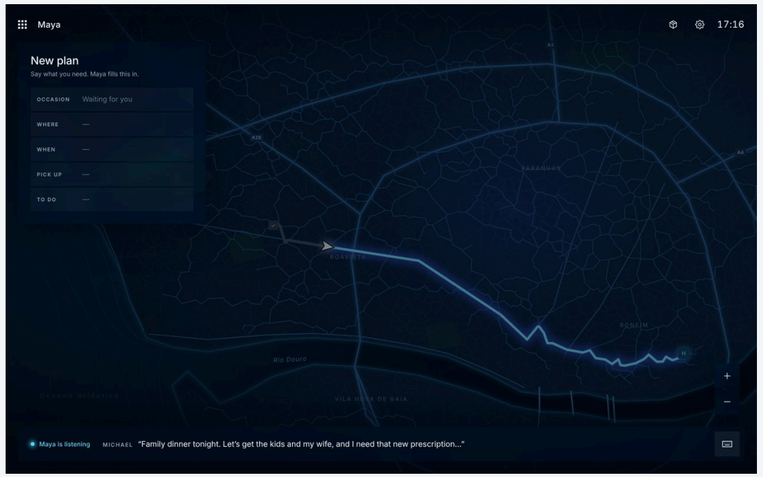
**SPOKEN**

| | |
|---|---|
| MICHAEL | “Hi Maya. Family dinner tonight. Let’s get the kids and my wife, and I need that new prescription from my messages as soon as possible.” |
| MAYA | “Where is the dinner?” |
| MICHAEL | “The usual Italian on Main Street.” |
| MAYA | “What time?” |
| MICHAEL | “Seven.” |
| TOUCH TWIN | Type in Maya’s bar (S05) and answer in the search field (S06). |

**PANORAMIC VISION · MAYA’S HALF**

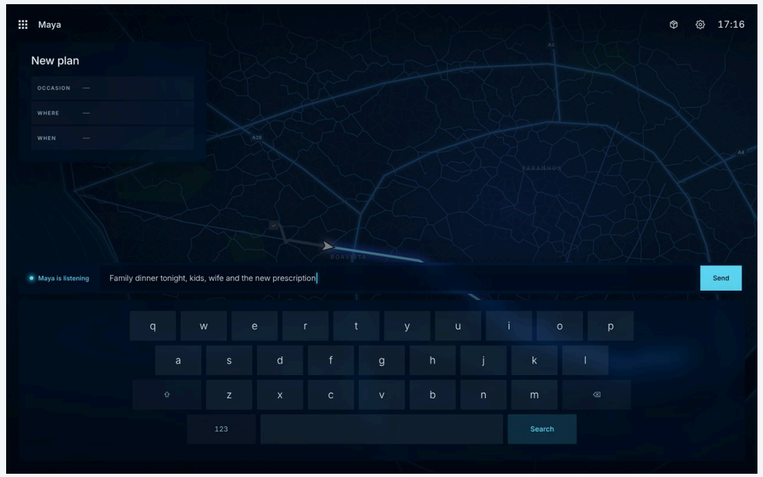



The presence dot brightens while Maya hears Michael. Home 17:48.



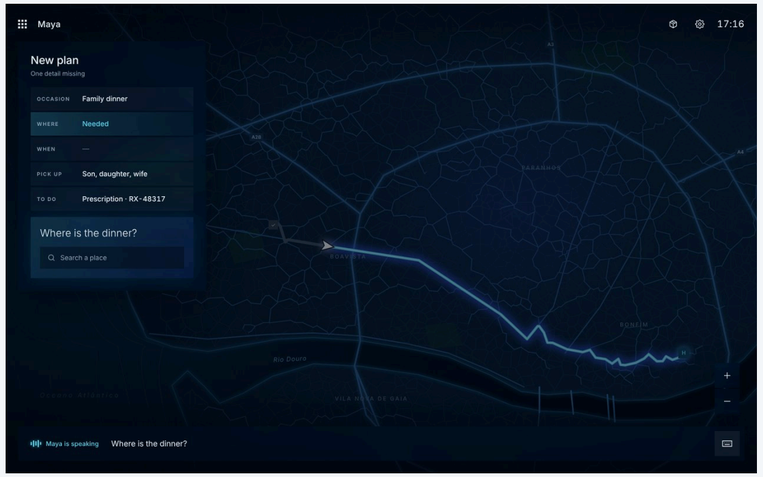

**S04** Briefing: listening

The “New plan” card: occasion, where, when, pick up, to do. It fills in as Michael talks.



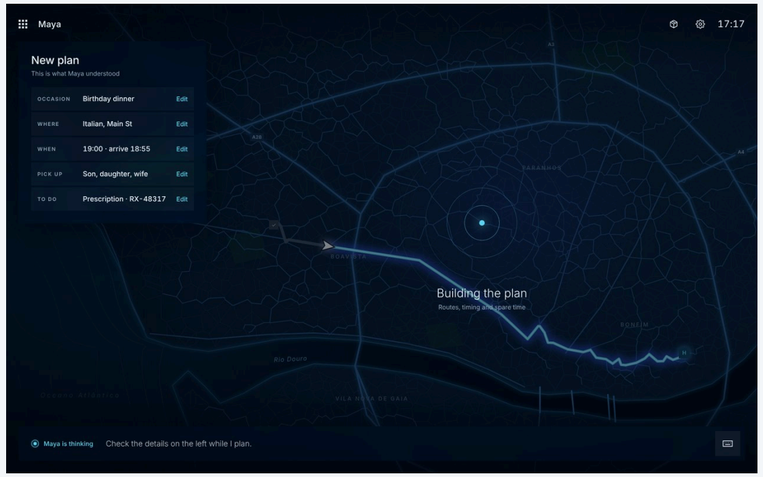

**S05** Briefing: keyboard

Touch twin: tap Maya’s bar to type. The keyboard takes the lower half; the card shrinks.



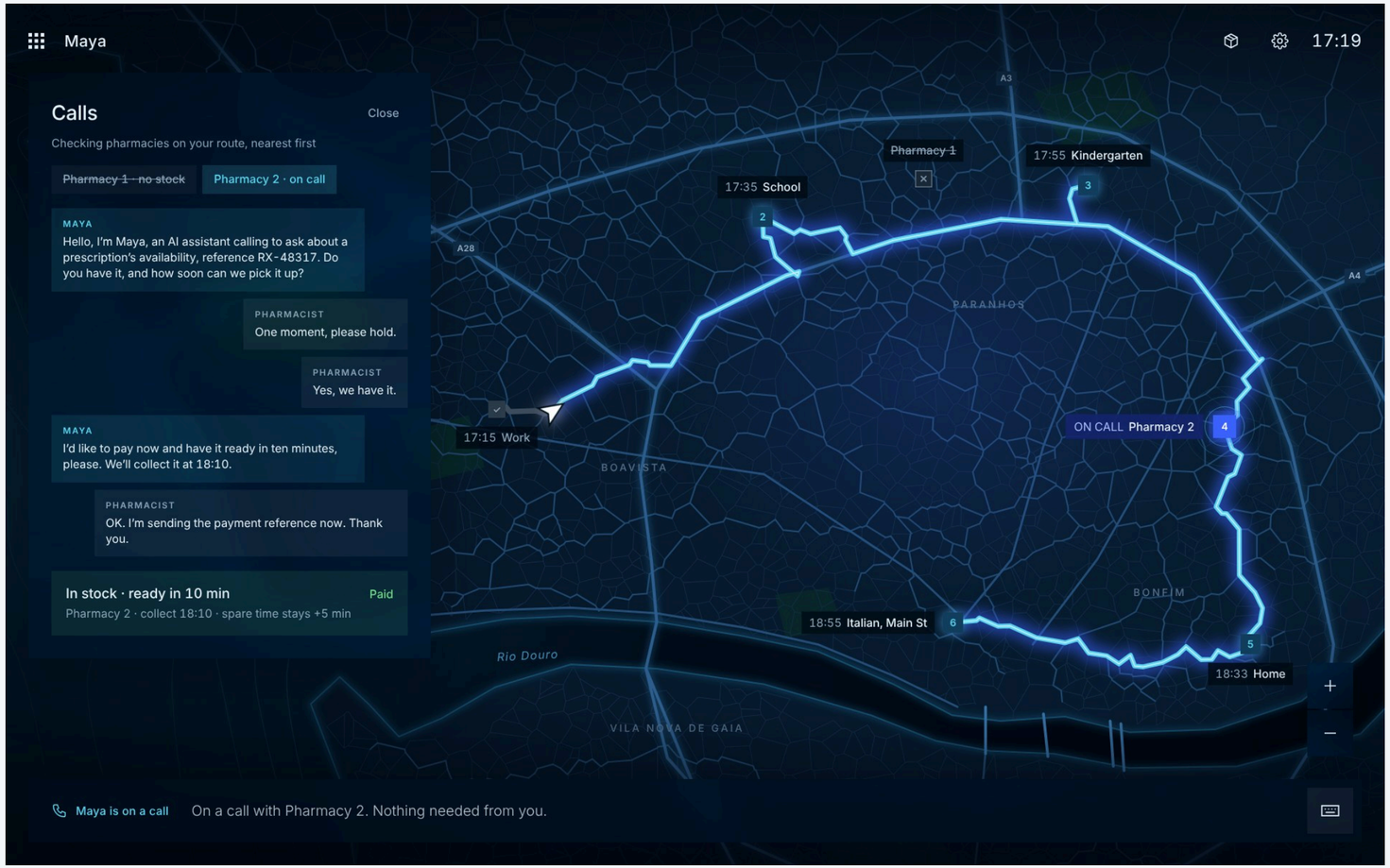

**S06** Briefing: Maya asks

Maya knows the pickups and the prescription. She asks only where, then when. No suggested answers.




**S07** Briefing: building the plan

“This is what Maya understood”, all still editable, while she builds the plan. No read-back.



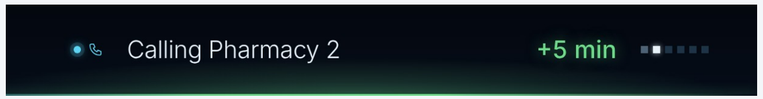

# Maya phones pharmacies on the route, nearest first. The cabin stays quiet.

## S08 Retailer calls

- Calls are silent in the cabin and transcribed live. Maya always says she is an AI.
- Pharmacy 1 has no stock and is struck through. Pharmacy 2 puts her on hold, then confirms.
- Paid from Maya Wallet, ready in 10 min, collected at 18:10. Spare time stays +5 min.
- On the map, Pharmacy 1 is crossed out and Pharmacy 2 marked "on call".
- The shared screen shows the prescription's reference number, never the medication.
- Maya uses a shop's own ordering system when it has one; phone calls are the fallback, because most local shops have none.

**TOUCH TWIN** Close hides the calls panel. Nothing is needed from Michael.

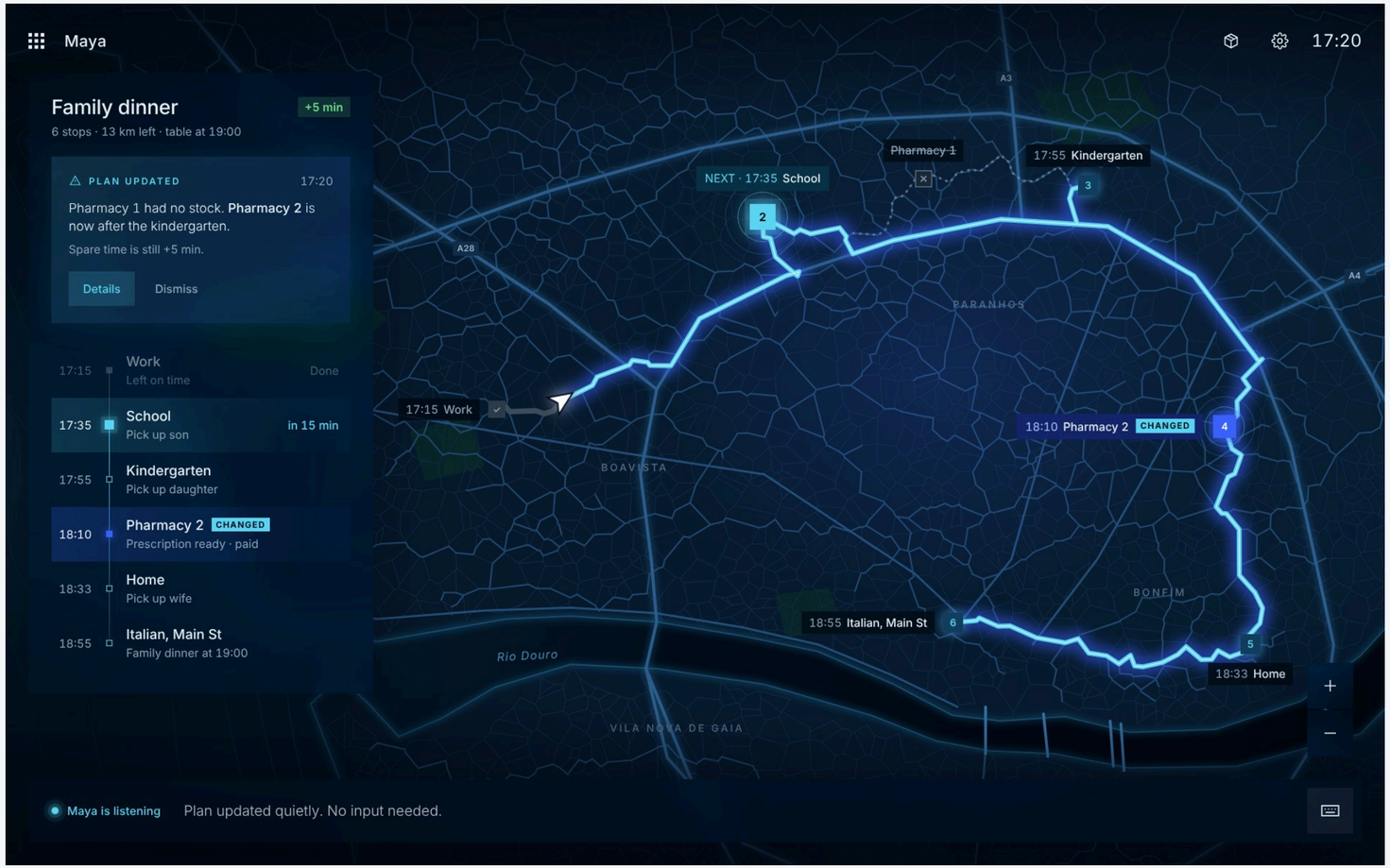
### PANORAMIC VISION · MAYA'S HALF




On a call: dot and phone. The plan now exists, so spare time appears: +5 min.

### SPOKEN

| | |
|---|---|
| MAYA | "Hello, I'm Maya, an AI assistant calling to ask about a prescription's availability, reference RX-48317." |
| PHARMACIST | "Yes, we have it." |
| MAYA | "I'd like to pay now and have it ready in ten minutes, please." |




# The world changed, so the plan changes quietly, with no input from Michael.

## S09 Plan change: automatic

- A change in the world, made silently: Pharmacy 1 had no stock, so Pharmacy 2 now comes after the kindergarten.
- No chime, no input. The banner says what changed and why, and stays until dismissed.
- Pharmacy 2 is marked CHANGED in the list and on the map; the old route stays dotted.
- Spare time is still +5 min.

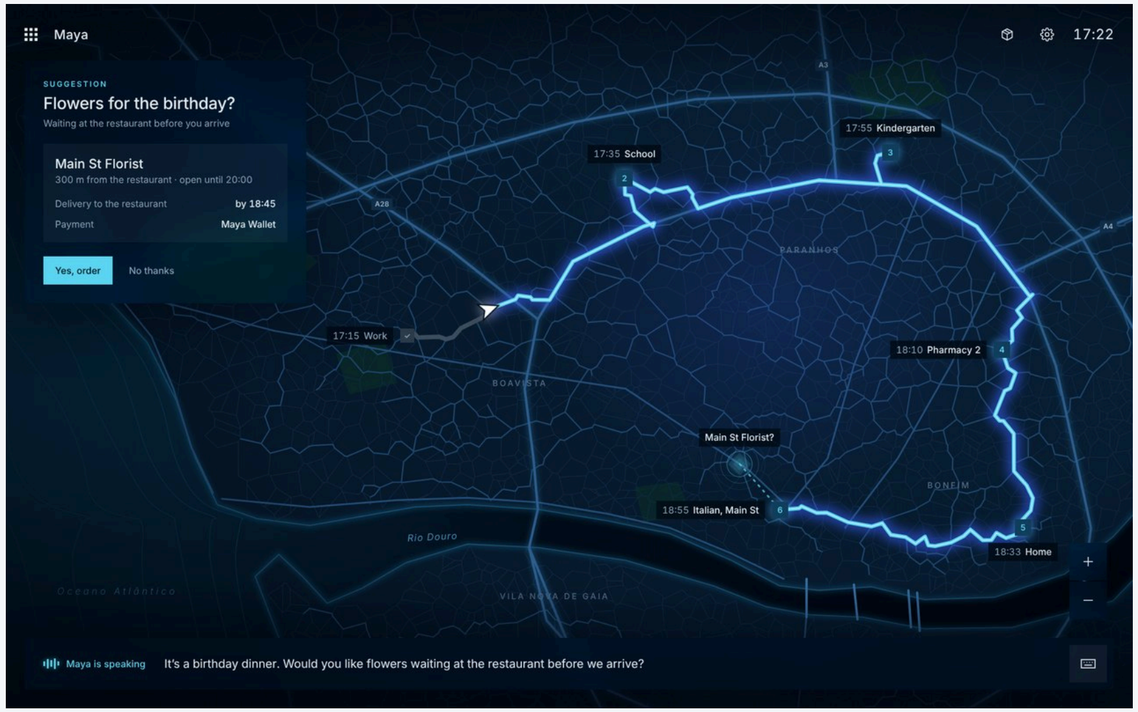
**TOUCH TWIN** Details reopens the calls; Dismiss closes the banner.

PANORAMIC VISION · MAYA'S HALF




A 4-second event: "Pharmacy changed +5 min". No chime; the line flashes white.

# A birthday in the calendar becomes one suggestion, then one call.

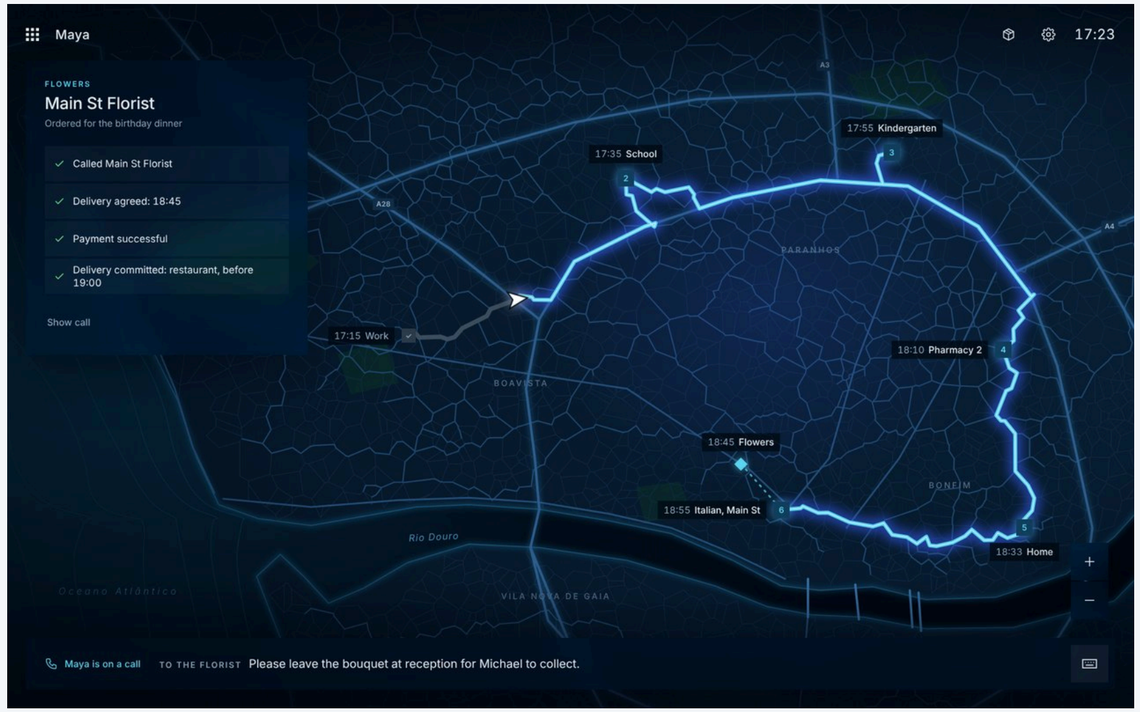



## S10 Gift suggestion

- Offered once the pharmacy is secured. One florist, no bouquet choices until Michael says yes.
- Main St Florist, 300 m from the restaurant: delivery by 18:45, paid with Maya Wallet.
- Until he answers, the florist is a dotted pin on the map.

SPOKEN

| | |
|---|---|
| MAYA | “It’s a birthday dinner. Would you like flowers waiting at the restaurant before we arrive?” |
| MICHAEL | “Yes. White roses.” |
| TOUCH TWIN | Yes, order or No thanks. |



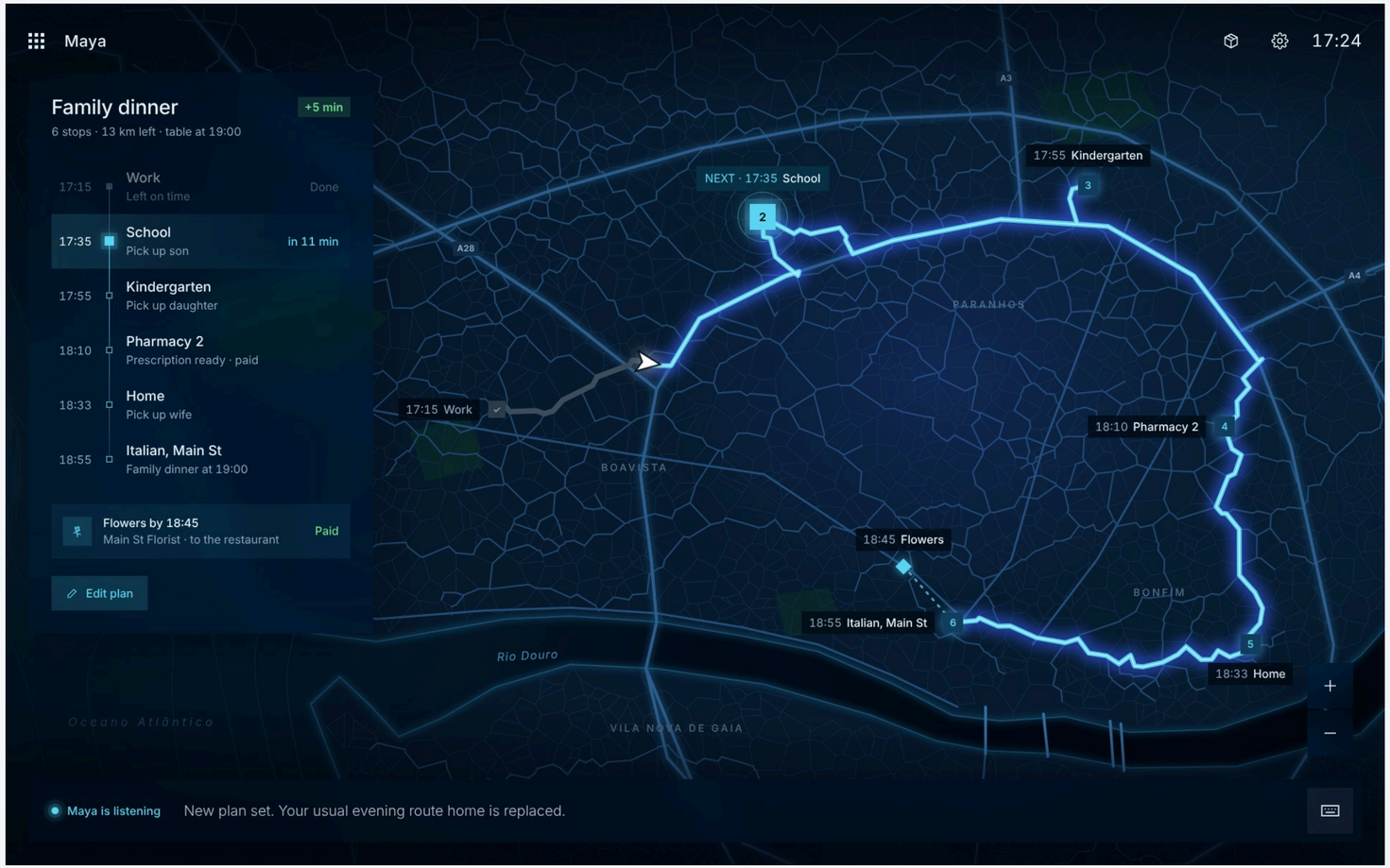

## S11 Florist order

- Four confirmed steps: called, delivery agreed for 18:45, payment successful, delivery committed before 19:00.
- Show call opens the transcript; Maya’s last words to the florist stay in her bar.
- The pin turns solid, with a dotted delivery line to the restaurant.

PANORAMIC VISION · MAYA’S HALF




Speaking: rings around the dot. School 17:35, +5 min.




PANORAMIC VISION · MAYA'S HALF




The resting state: next stop, spare time and one square per stop.

SPOKEN

MAYA'S BAR "New plan set. Your usual evening route home is replaced."

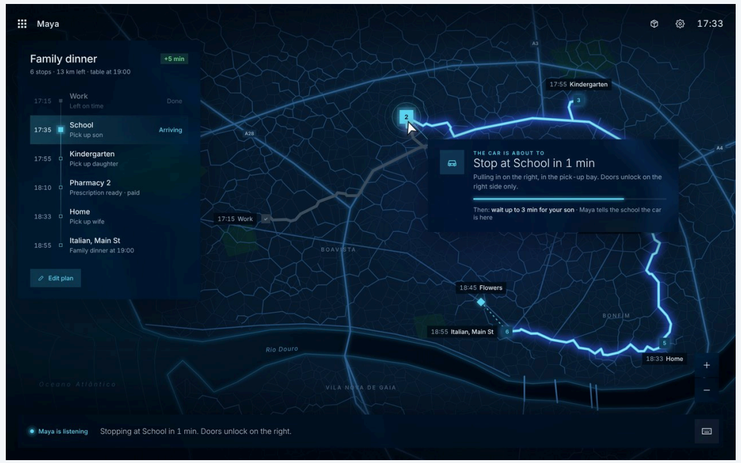
# The route home is replaced by the plan. No speech: the screen is the confirmation.

## S12 Plan overview

- Family dinner: 6 stops, 13 km left, table at 19:00, and +5 min of spare time at the top.
- The stop timeline: Work done and greyed, School next with "in 11 min".
- Flowers by 18:45, paid, in their own row; Edit plan below.
- The whole trip on the map: the route ahead glows, the travelled route is grey, the next stop has sonar rings.
- The car never stopped moving: the plan simply takes over from "Going home".
- Maya checks the range as she builds the plan: about 15 km of driving, well within it.

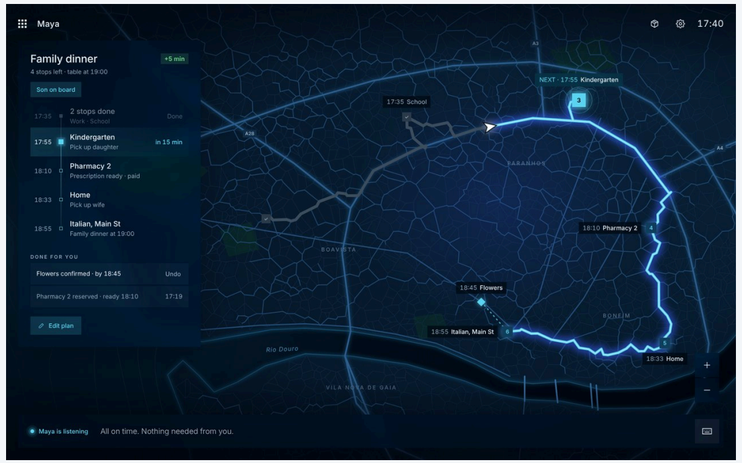
TOUCH TWIN Tap a stop, Edit plan (S15), zoom. By voice: "What's next?"

# The car says what it is about to do. Maya keeps a ledger of what she did.



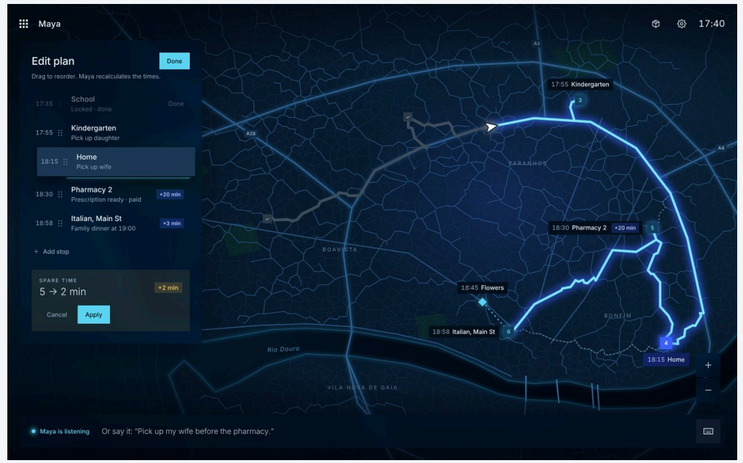

## S13 Vehicle intent

- About a minute before each stop: where the car pulls in and which doors unlock.
- Then: wait up to 3 min; Maya tells the school the car is here.
- The stop row reads “Arriving”.




## S14 Active trip

- “Son on board”; finished stops fold into one row.
- “Done for you”: the last two actions, with Undo on the latest only.
- Edit plan is always one tap away.



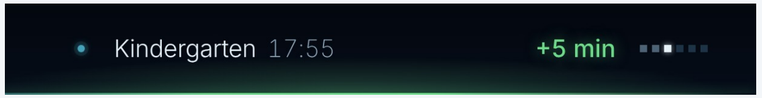

## S15 Edit plan

- A branch: drag to reorder. The ripple shows every later time, +20 min, +3 min.
- Spare time 5 → 2 min, in amber. Nothing changes until Apply.
- The map previews the new route; the old one is dotted.

### SPOKEN

| | |
|---|---|
| MAYA’S BAR | “Stopping at School in 1 min. Doors unlock on the right.” |
| MAYA’S BAR | “All on time. Nothing needed from you.” |
| TOUCH TWIN | Edit plan, or say “Pick up my wife before the pharmacy.” |

### PANORAMIC VISION · MAYA’S HALF



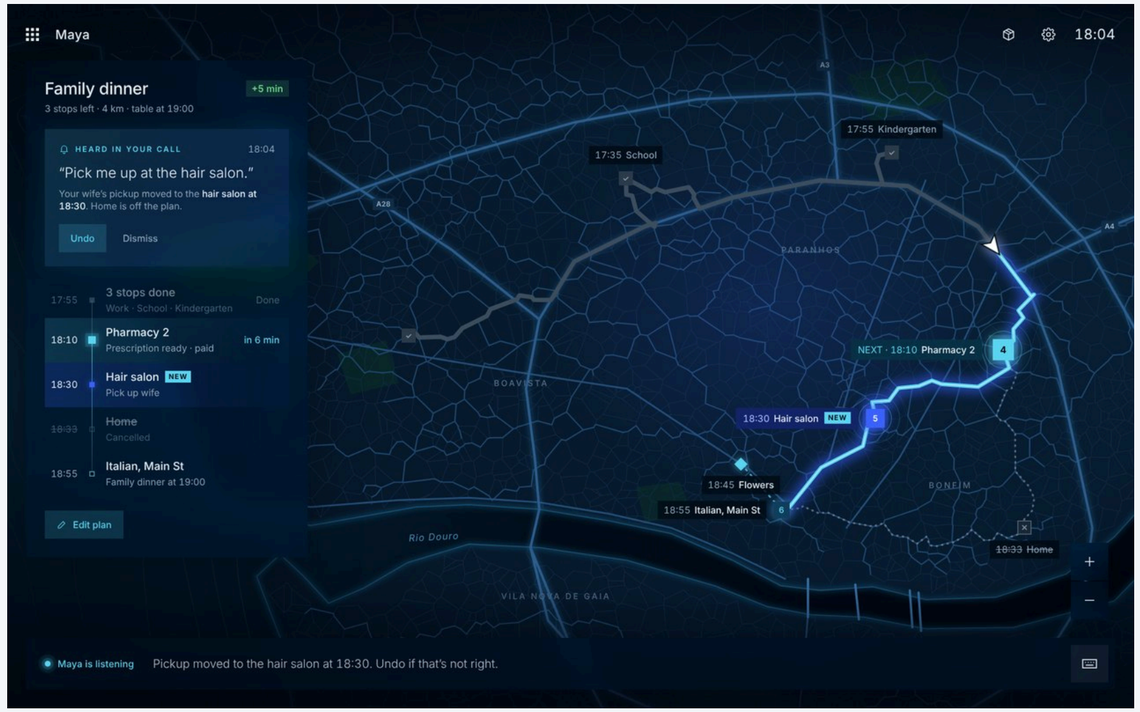

The next stop updates as stops complete: Kindergarten 17:55, +5 min.

# Maya hears “pick me up at the hair salon” and moves the stop, with a chime and Undo.




## S16 Incoming call

- A call card over the map: Wife, Mobile, Answer and Decline. The plan stays visible behind it.
- Maya listens quietly; the other person hears a short notice that she is listening.



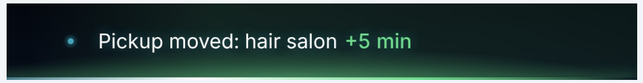

## S17 Plan change: ambient

- “Heard in your call”: the phrase, what changed, then Undo and Dismiss. No countdown.
- Hair salon marked NEW at 18:30; Home crossed out; the old route dotted.
- One soft chime, no speech. If Maya were unsure, she would ask instead.

### SPOKEN

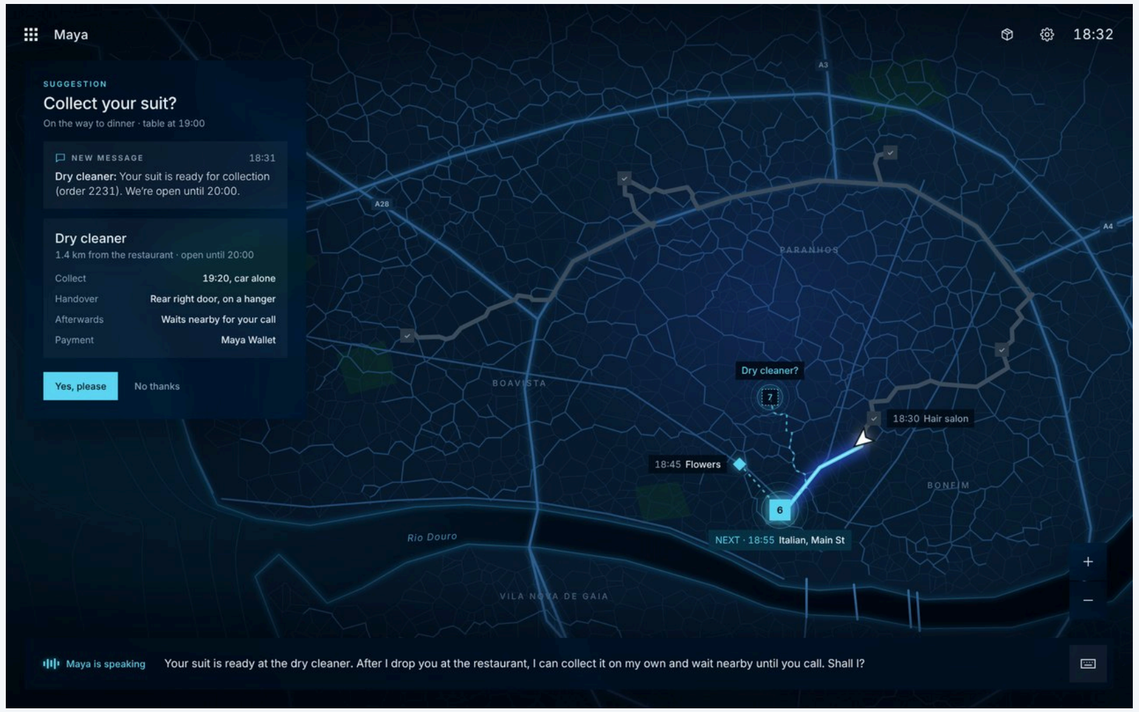
| | |
|---|---|
| WIFE | “Change of plan: pick me up at the hair salon, not at home.” |
| MICHAEL | “Sure, we’ll be there.” |
| TOUCH TWIN | Undo, or change the pickup in Edit plan (S15). |

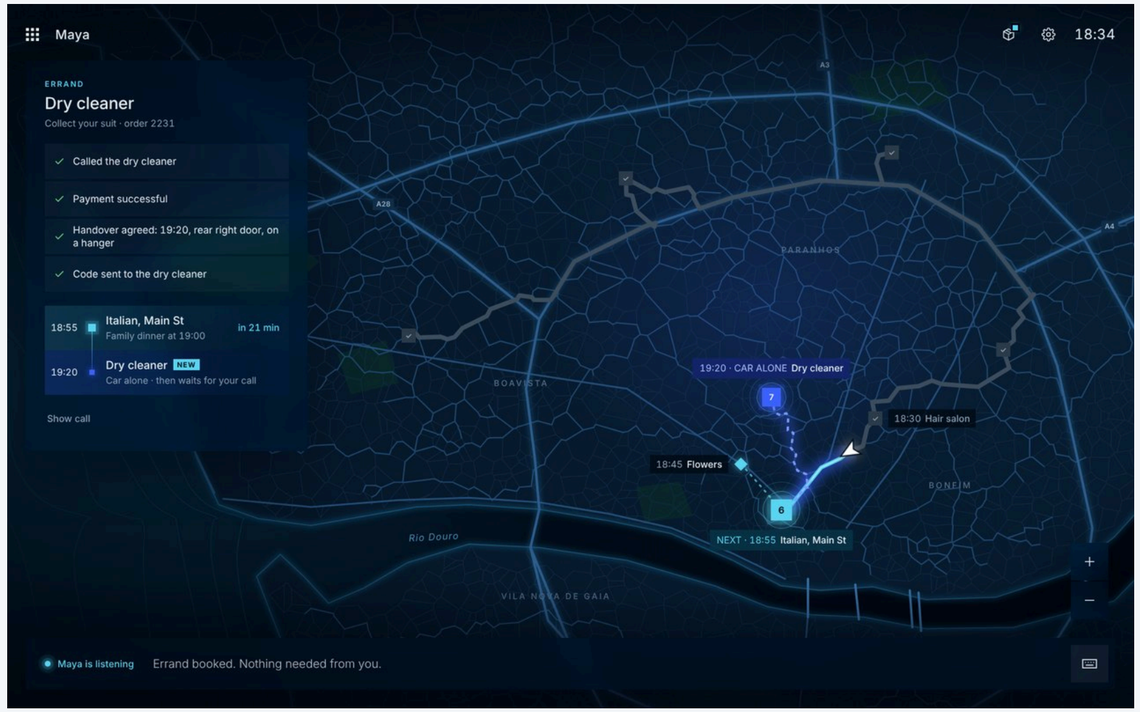
### PANORAMIC VISION · MAYA’S HALF




A 4-second event: “Pickup moved: hair salon +5 min”. The line flashes white.

# A message from the dry cleaner becomes an errand for the empty car.




## S18 Errand suggestion

- The dry cleaner's message is shown, not read aloud.
- The offer: 19:20, car alone, rear right door, then it waits nearby for his call; paid with Maya Wallet.
- The dry cleaner is a suggested pin until Michael answers.

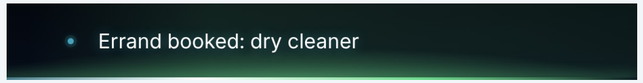
## S19 Errand booked

- Booked only once the handover is agreed: called, paid, handover agreed, code sent.
- A new plan row after the restaurant: 19:20 Dry cleaner, car alone, then waits for his call.
- A dot on the Errands icon while an errand is booked or running.

SPOKEN

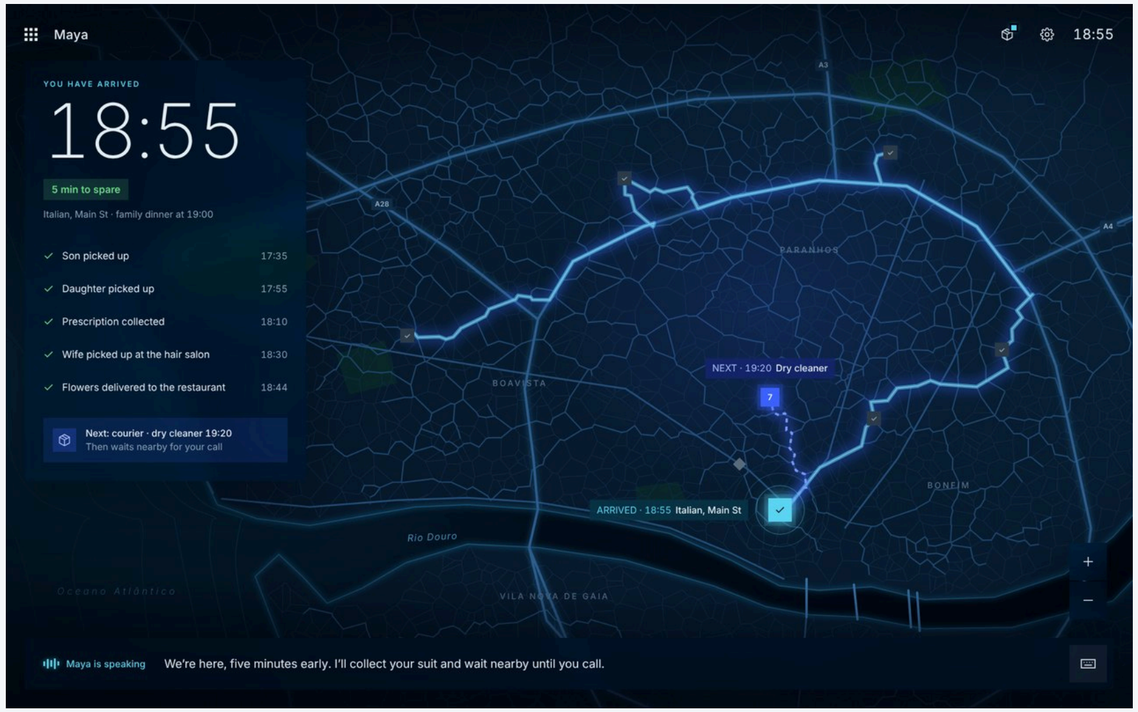
| | |
|---|---|
| MAYA | "Your suit is ready at the dry cleaner. After I drop you at the restaurant, I can collect it on my own and wait nearby until you call. Shall I?" |
| MICHAEL | "Yes, please." |
| TOUCH TWIN | Yes, please or No thanks. |

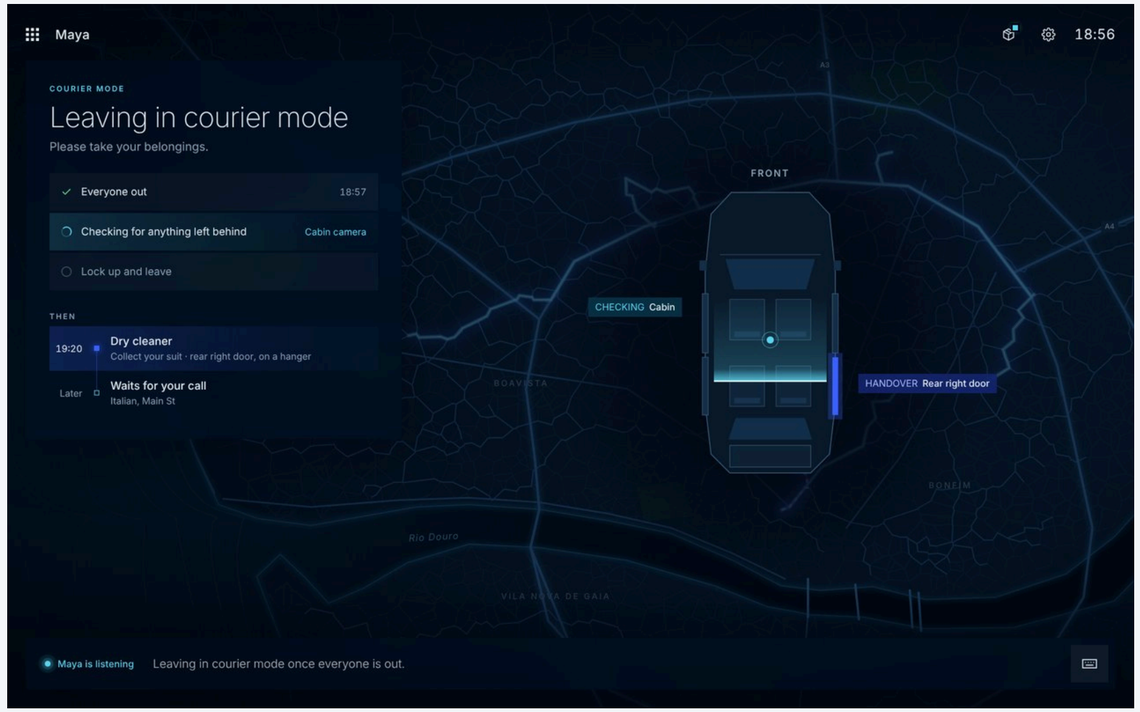
PANORAMIC VISION · MAYA'S HALF




"Errand booked: dry cleaner" for 4 seconds. Spare time is unchanged.

# Five minutes early. Then the car checks it is empty and leaves on its errand.




## S20 Arrival report

- 18:55 and 5 min to spare, readable in one glance.
- Everything done, with times: son, daughter, prescription, wife at the hair salon, flowers delivered 18:44.
- Next: courier, dry cleaner 19:20, then the car waits nearby for his call.

## C1 Leaving in courier mode

- "Please take your belongings."
- Everyone out, then the cabin camera checks for anything left behind, then lock up and leave.
- A top view of the car shows the cabin check and the handover door.

SPOKEN

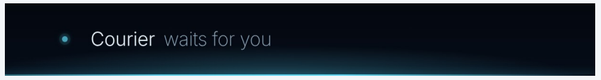
MAYA "We're here, five minutes early. I'll collect your suit and wait nearby until you call."

PANORAMIC VISION · MAYA'S HALF

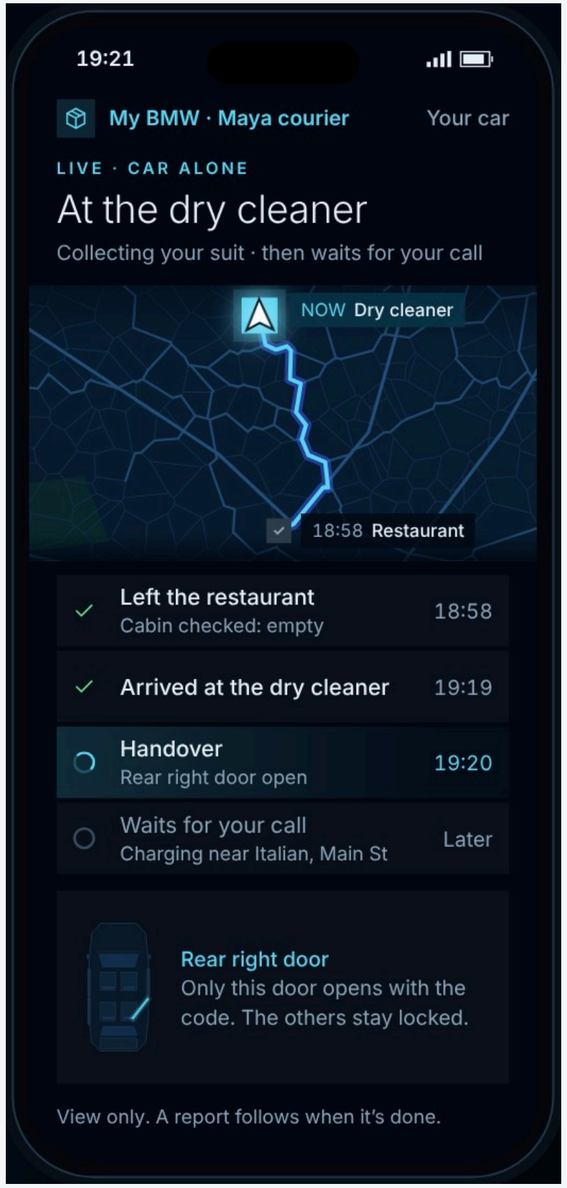


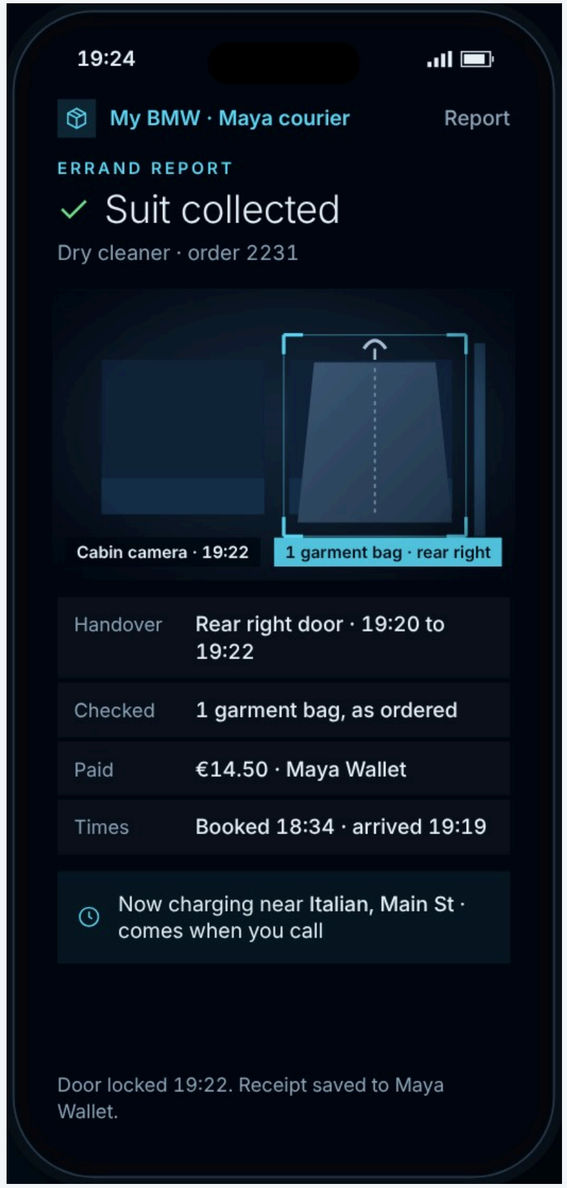

Arrived 18:55, every square filled, +5 min.

THEN




"Courier · waits for you" while the family gets out.




# Michael follows the run on his phone, view only, and gets a report.

## C2 Phone: courier run

- Live, with the car alone, in the My BMW app: the map and the steps. Left the restaurant 18:58, arrived 19:19, handover 19:20, then it waits for his call.
- Which door is in use: only the rear right door opens with the code; the others stay locked.
- View only: no controls and no voice.

## C3 Phone: errand report

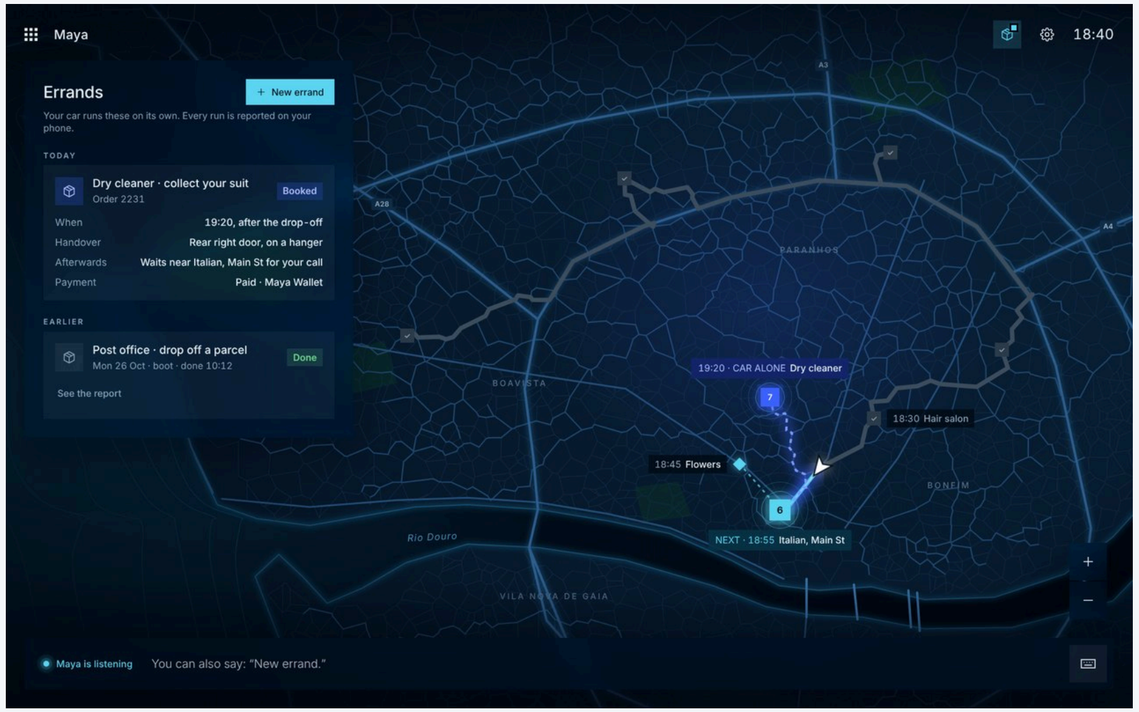
- A still from the cabin camera: 1 garment bag, rear right.
- Handover 19:20 to 19:22, checked as ordered, paid €14.50, booked 18:34, arrived 19:19.
- Now charging near the restaurant until he calls. Door locked 19:22, receipt saved to Maya Wallet.

### WHERE THE STORY ENDS

The car drives to a charging station near the restaurant. Charging is not an errand, so it never appears in the plan. The same car, with the suit inside, waits there until Michael calls it.

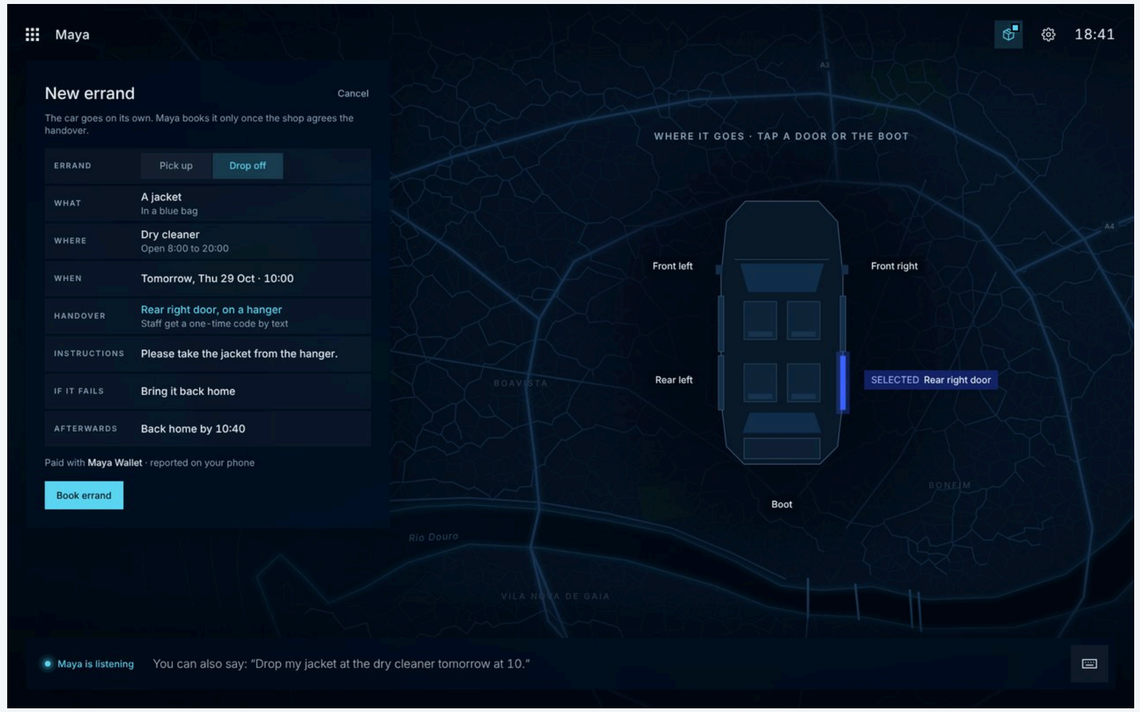
DISPLAYS Off: the cabin is empty, so the central display and Panoramic Vision are off.

# Errands can also be booked by hand, from the box icon beside Settings.




## C4 Errands

- Today: the dry cleaner, Booked, with when, handover, what happens afterwards and payment.
- Earlier: Post office, drop off a parcel, Mon 26 Oct, boot, Done, with its report.
- New errand at the top.




## C5 New errand

- Pick up or Drop off; what, where, when.
- Handover: tap a door or the boot on a top view of the car. Staff get a one-time code by text.
- Instructions, what to do if it fails, what the car does afterwards; then Book errand.

SPOKEN

VOICE TWIN "Drop my jacket at the dry cleaner tomorrow at 10."

# Tokens verified against automotive guidance.



**Abyss**
Background

**Glass**
Panels, 30–60% over the map

**Lumen**
Primary accent

**Cobalt**
Glow, new items, errands

**Green**
More than 4 min spare

**Amber**
0 to 4 min spare

**Red**
Late, takeover

**Text**
Primary

**Muted**
Secondary

## TYPE · INTER

| | | |
|---|---|---|
| **Display** | 128 / ExtraLight | Arrival time (S20) |
| **Hero** | 44–60 / Light | Sign-in, call card, courier title |
| **Large** | 24–30 / Light | Clock, heard phrase, intent |
| **Title** | 28 / Regular | Panel titles |
| **Body** | 19–20 / Regular | Stops, Maya's captions |
| **Secondary** | 16–17 / Regular | Times, details, notes |
| **Label** | 13 / SemiBold caps | Kickers and row labels |
| **Badge** | 12 / SemiBold caps | NEW, CHANGED, CHEAPEST |

Dark first: the night cabin is the hard case, and light can carry meaning. Panoramic Vision is always dark. Material 3 and Android for Cars are references for sizes and behaviour only.

## RULES

- **Sharp corners, no borders.** Fill, glass and spacing group things, never lines.
- **Glass over a full-bleed map.** Panels at 30 to 60% with a 24 px blur, so the map is always there.
- **76 × 76 px tap areas.** Visual sizes stay (48 to 64 px), but every tap area meets the Android for Cars minimum of 76 dp, at about 1 px per dp.
- **16 px minimum text.** Except uppercase labels (13 to 14 px) and small badges (12 px).
- **About 15″, 1920 × 1200, at about 70 cm.** A 16:10 central display, like today's large infotainment screens.
- **Colour is never alone.** Spare time always has a number; late adds a minus sign and a warning mark.
- **One sound.** A soft chime, only for changes heard in conversation.
- **Motion with a job.** Panoramic Vision events last 4 seconds; Maya's presence pulses slowly; nothing loops while she speaks.

# Component inventory

38 components, drawn with the same code as the screens · every tap area is at least 76 × 76 px

### Top bar — CHROME
App launcher · Maya wordmark · Errands button with badge · Settings · clock. Clock only before sign-in
All central display screens

### Icon button — CHROME
Default · active · with badge. 48 to 64 px visual, 76 px tap area
All central display screens

### Maya’s bar — CHROME
Listening · thinking · speaking · on a call. Caption in one or two lines, with speaker label. Keyboard button; text-entry mode
S02 to S20, C1, C4, C5, V1, V2

### Zoom control — CHROME
Zoom in · zoom out
Map screens

### Route line — MAP
Live · travelled · previous plan · delivery · courier · suggested errand · traffic
S02 to S20, C1, C4, V2

### Stop pin — MAP
Upcoming · next with sonar rings · done · new · cancelled · suggested · arrived
S02 to S20, C4, V1, V2

### Delivery pin — MAP
Default · suggested · done
S10 to S20

### Car marker — MAP
Moving, with heading
S02 to S19, C4, V2

### Map label — MAP
Upcoming · next · new · done · cancelled · suggested. Place names and road shields
S02 to S20, C4, V2

### Panel — CONTENT
Narrow · wide · form. Glass at 30 to 60 percent, 24 px blur, no border
S02 to S20, C1, C4, C5, V2

### Spare-time pill — CONTENT
Green · amber · red with minus and warning mark · large
S09, S12 to S17, S20, V2

### Stop row — CONTENT
Upcoming · next · arriving · done · summary · new · changed · cancelled · dragging
S09, S12 to S15, S17, S19, C1, V2

### Briefing slot — CONTENT
Empty · asked · filled · editable
S04 to S07

### Suggestion card — CONTENT
Florist · errand, with key–value rows and a quoted message
S10, S18

### Change banner — CONTENT
Automatic · heard phrase with chime and Undo · running late with fixes
S09, S17, V2

### Step list — CONTENT
Done · running · waiting
S11, S19, C1, C2

### Ledger row — CONTENT
Latest with Undo · older with time
S14, S16

### Call transcript — CONTENT
Maya · other party, hold timer, outcome card
S08

### Info row — CONTENT
Icon tile, title, detail and status
S12, S20

### Chip — CONTENT
Default · active · failed · state
S02, S03, S08, S14, S16

### Badge — CONTENT
New · changed · cheapest · booked · done · limit
S03, S09, S17, S19, C4, V2

### Button — INPUTS
Primary · secondary · text · destructive · call
All screens

### Segmented control — INPUTS
Two or three options
S03, C5

### Toggle — INPUTS
On · off
S03

### Text field — INPUTS
Search · free text with caret · with Send
S05, S06

### Keyboard — INPUTS
Letter · modifier · space · action key
S05

### Form row — INPUTS
Label and value · with hint · selected
C5

### Option row — INPUTS
Recommended · default, each with an action
V2

### Incoming call — OVERLAYS
Answer · decline
S16

### Intent card — OVERLAYS
Action, detail, progress and what comes next
S13

### Takeover alert — OVERLAYS
Countdown ring, red, Level 3/4 variant only
V1

### Sign-in — OVERLAYS
Face scan · PIN or password · switch profile
S01

### Processing — OVERLAYS
“Building the plan” with sonar rings
S07

### Arrival report — OVERLAYS
Large time, pill, what was done, next step
S20

### Car top view — OVERLAYS
Default · selected compartment · door open · cabin scan
C1, C2, C5

### Phone run — PHONE AND PANORAMIC VISION
App bar, live map card, steps: done · running · waiting
C2, C3

### Camera still — PHONE AND PANORAMIC VISION
Detection frame, camera label, result chip
C3

### Panoramic Vision band — PHONE AND PANORAMIC VISION
Presence · next stop · spare time · progress · event · horizon line in cyan, green, amber, red · offline · off
All Panoramic Vision states

# Panoramic Vision

One band, always dark and not touchable, that shows nothing unique. Never on it: transcripts, maps, editing, anything long or interactive, medication names, personal data.




**1 Automotive half**
Speed, cabin temperature and range. Always on the left, always the same.

**2 Presence**
A small dot: listening, hearing you, speaking (rings), thinking (hollow), on a call (phone).

**3 Next stop**
Name and planned time. Replaced by an event for 4 seconds.

**4 Spare time**
The figure and the line share one colour: green, amber or red.

**5 Progress**
One square per stop: grey done, white current, dark to come.

**6 Horizon line**
Cyan under the car data, the spare-time colour under Maya. Flashes white on events.

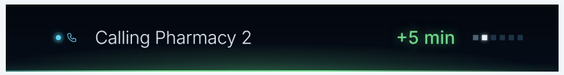
## STATES THROUGH THE STORY · MAYA'S HALF

### S02 Signed in




No plan yet: neutral line, no spare time.

### S08 Pharmacy call




On a call. The plan exists, so spare time appears.

### S09 Event · plan changed



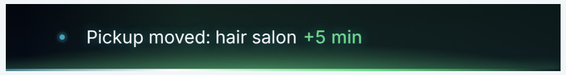

A change in the world: no chime, the line flashes white for 4 s.

### S12 Plan running



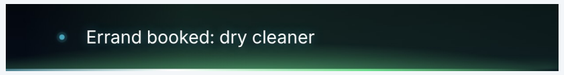

Next stop, spare time, one square per stop.

### S17 Event · ambient change




Heard in the call: the chime sounds once, the line flashes.

### S19 Event · errand booked



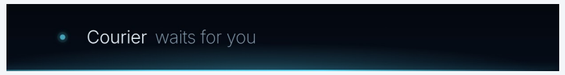

Agreed with the shop. Spare time unchanged.

### S20 Arrival




Every square filled; the final spare time.

### C1 Courier hand-off




What happens next: the car waits for Michael's call.

### C2 Off · cabin empty



Nobody inside: the whole band is off.

### Amber Low spare time · 0 to 4 min



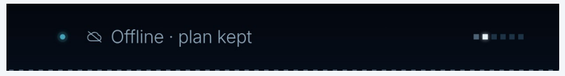

The number is always shown; the line turns amber.

### V2 Red · below 0



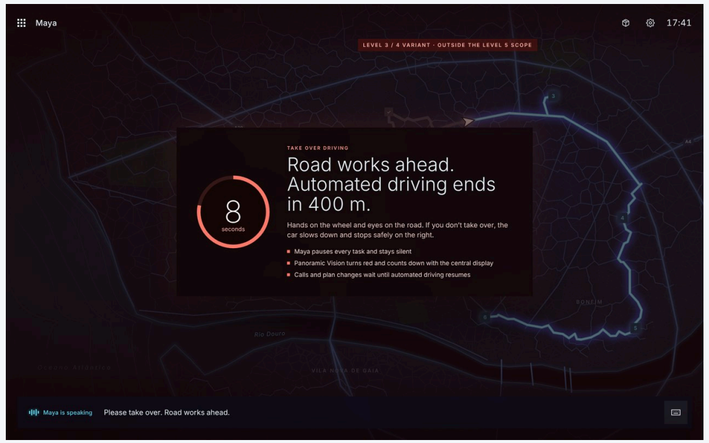

A minus sign and a warning mark, never colour alone.

### Edge 8 Offline · plan kept



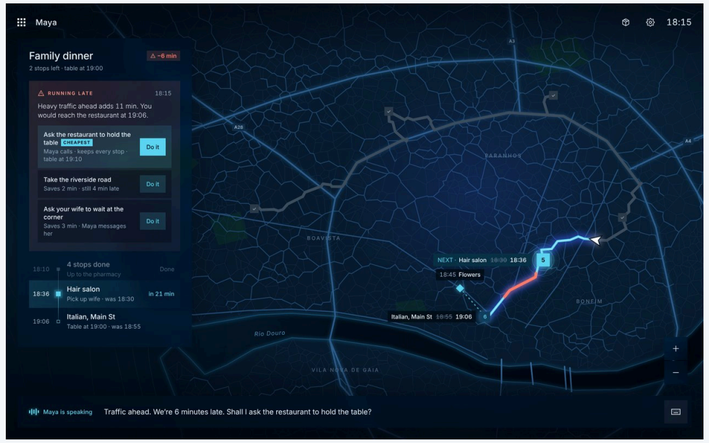

A grey dotted line; the plan keeps running.

# Seventeen ways the evening could go wrong.




**V1** Takeover request

Level 3 or 4 only, outside the Level 5 scope: an 8-second countdown on the central display and Panoramic Vision; Maya pauses every task and stays silent.




**V2** Running late · edge case 2

−6 min with a warning mark. Three fixes, cheapest first, each with Do it. Maya speaks once, because the promise is at risk.

| # | EDGE CASE | EXPECTED BEHAVIOUR |
|---|---|---|
| 1 | No pharmacy on the route has stock | Widen the search, show it on screen; ask only if the detour breaks spare time |
| 2 | Spare time goes negative | Fixes, cheapest first, such as holding the table (V2); Panoramic Vision turns red |
| 3 | Ambient false positive: a child says “let’s go to the park” | Act only when confident, otherwise ask; chime, heard phrase and Undo |
| 4 | The chime is missed | The change stays on screen until dismissed, and in the ledger; Panoramic Vision repeats it once |
| 5 | The florist can’t deliver before 19:00 | Try the next florist; otherwise offer a pickup stop |
| 6 | A pharmacist refuses an AI caller | Next pharmacy, with the refusal logged |
| 7 | A child is not at the pickup point | Alert on the central display, a wait window, a call to the school |
| 8 | Connectivity loss | Keep the current plan; Panoramic Vision shows Offline; edits queue |
| 9 | Dinner time or place changes by message | Same pattern as the wife’s call: chime and Undo |
| 10 | Several people speak | Commands only from the signed-in user; other speech is context |
| 11 | Shop not ready, or nobody comes to the car | Wait within a set window, then leave; the report says why, and the errand returns to Errands |
| 12 | Wrong item loaded, or more than one | The camera checks before the door locks; Maya texts the shop and waits within the window |
| 13 | Someone or something left in the cabin | A person or a pet stops courier mode; a bag gets “Please take your belongings” and a note on the phone |
| 14 | The car is needed before the errand is done | Michael calls it; the same car comes from charging, or finishes the handover first |
| 15 | The errand runs late (traffic, a slow handover) | Tell Michael early on the phone; if he calls the car, it shows when it can reach him |
| 16 | Wrong door, or the code is shared | The code opens only the agreed door, once, at the agreed place and time |
| 17 | Item not allowed: too big, an animal, needs a signature | Declined while booking, before anything is promised |

Cases 11 to 17 cover courier mode. Cases 1, 3 and 4 also appear in the prototype.

# Riskiest first: does ambient intent feel helpful, or creepy? Would people send an empty car?

## TESTS

| TEST | METHOD | SUCCESS SIGNAL | CHECKS |
|---|---|---|---|
| **Ambient-intent false positives** | Scripted audio clips, true changes and decoys; measure Undo and reaction | Few false positives; chime and phrase understood | A3, A9 |
| **Courier trust test** | Clickable courier flow, S18 to C3, 5 participants: what would they send, what would stop them? | 4 of 5 would send it for an everyday errand | A14 |
| **Glance test** | 5-second exposure: “What’s next? How much spare time?” | 80% or more correct | A8, A10 |
| **Replan legibility** | Show a change, ask “what changed?” | Understood in under 5 s | A9 |
| **Touch-only run** | Brief and edit a plan without speaking | Same completion as voice | A11 |
| **Wizard of Oz briefing** | A person plays Maya’s voice, 5 participants (if I had a month) | No confusion; count the questions | A1, A6 |

The glance, legibility, touch-only and courier trust tests can run remotely on the clickable prototype. Scripted calls to real shops (A4, A13) and the Wizard of Oz session are on the one-month list.

## ASSUMPTIONS · HIGH RISK IN RED

| | |
|---|---|
| **A1** | Users accept “do first, show clearly, undo easily” |
| **A2** | Maya can predict the destination and learn the day from habit and data |
| **A3** | Listening to cabin talk and calls is welcome; false positives are cheap |
| **A4** | Retailers cooperate with a self-identified AI caller |
| **A5** | Automatic payments from Maya Wallet, up to a limit, are acceptable |
| **A6** | Users accept as many questions as needed, if each is necessary |
| **A7** | A suggestion tied to a known occasion is welcome |
| **A8** | Spare time gives at-a-glance confidence |
| **A9** | Chime, heard phrase and Undo are understood in a noisy cabin |
| **A10** | A passive Panoramic Vision carries enough at a glance |
| **A11** | Voice and touch parity is achievable for every task |
| **A12** | On-time arrival is the main success criterion |
| **A13** | Shops will hand items to an empty car that opens one door with a code |
| **A14** | People will send their empty car, and accept a cabin camera |
| **A15** | An empty Level 5 car may legally run errands |
| **A16** | Michael’s phone data (messages, calendar) is available to the car, despite platform limits |

## SUCCESS METRICS · ILLUSTRATIVE TARGETS

| | |
|---|---|
| On-time arrival, within the 5 minutes of spare time | **95% or more** |
| Replans that keep spare time at or above zero | **90% or more** |
| Questions Maya could have answered herself | **Under 10%** |
| Undo rate on automatic and ambient changes | **Under 10%** |
| Time from a change to a visible adapted plan | **3 s or less** |
| Proactive suggestions accepted | **50% or more** |
| “Feels like a great assistant” | **4 / 5 or more** |
| Courier errands completed without Michael stepping in | **90% or more** |
| After a courier run, the car arrives within the time shown when called | **95% or more** |
| Actions moved from Ask first to Act within a month | **Rising** |

# What each choice gains, what it costs, and what I would do with a month.

| DECISION | GAIN | COST |
|---|---|---|
| Level 5 only | One calm, trip-level experience | No driver-state design; V1 sketches Level 3 or 4 |
| Full trust: Maya acts, pays and listens | Feels like an executive assistant | No approval before calls, or payments under the limit; trust lives in S03 and Undo |
| Maya speaks rarely | A quiet cabin | Voice-only users get less status; captions and Panoramic Vision help |
| Ambient intent: act when sure, ask when unsure | Plans follow life without being managed | False positives, answered by chime, heard phrase and Undo |
| Silent retailer calls, shown as transcripts | Silence in the cabin | Michael can read what Maya promised, not hear it |
| Banners stay until dismissed | Nothing is lost | Clutter if changes pile up |
| Undo on the latest ledger row only | A calm ledger, one clear action | Older actions are slower to reverse |
| Edit plan with a ripple preview, then Apply | Informed edits | An extra step for a quick change |
| A passive Panoramic Vision with nothing unique | No second interactive surface | It can never be the only channel |
| One layout: plan left, flat map | Learnable and consistent | No right-to-left variant, no 3D context |
| Courier: no handoff, no commitment | Errands don't fail at the door | Fewer errands qualify |
| Courier: one door, one code, a cabin camera | The rest of the car stays closed | A camera in the cabin; the item must fit the compartment |
| A solo exercise, no user research | Speed | A1 to A16 are untested |

## IF I HAD A MONTH

- Wizard of Oz sessions and a study of real family trips
- Scripted calls to real shops, for AI calls and empty-car handovers
- Other errand types beyond pick-ups and drop-offs
- Responsive layouts: each screen size designed, not scaled
- Trust that grows over time, a full activity log, wallet top-ups and statements
- Multi-passenger conflicts and escalation to a human
- Accessibility research with touch-only and voice-only users
- A regulation review for empty cars (A15)
- When Maya is unsure about a payment: ask first, or act and show Undo (leaning towards Undo, below a limit)
- Screens for later: the app launcher's contents, creating a new profile, idle or standby, guest mode
- A shared digital key for handovers, instead of a typed code
- The 3D head-up display
- Driving preferences, such as "prioritise speed over cost"

THANK YOU

# Thank you.

“This is definitely the most fun challenge I’ve done for a job application.”

Aníbal Oliveira · Senior Product Designer

Don’t forget to visit Maya’s official page:

oliveiraanibal.github.io/maya

The entry page, the working artifacts and the clickable prototype.